How to Read a Scientific Article

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Reading a scientific article is a complex task. The *worst* way to approach this task is totreat it like the reading of a textbook—reading from title to literature cited, digesting every word along the way without any reflection or criticism. Rather, you should beginby skimming the article to identify its structure and features. As you read, look for the author's main points. Generate questions before, during, and after reading. Draw inferences based on your own experiences and knowledge. And to really improve understanding and recall, take notes as you read. This handout discusses each of these strategies in more detail.

1. Skim the article and identify its structure.

Most journals use a conventional IMRD structure: An abstract followed by Introduction,Methods, Results, and Discussion. Each of these sections normally contains easily recognized conventional features, and if you read with an anticipation of these features, you will read an article more quickly and comprehend more.

Features of Abstracts

Abstracts usually contain four kinds of information:

- purpose or rationale of study (why they did it)
- methodology (how they did it)
- results (what they found)
- conclusion (what it means)

Most scientists read the abstract first. Others—especially experts in the field—skip rightfrom the title to the visuals because the visuals, in many cases, tell the reader what kinds of experiments were done and what results were obtained. You should probably begin reading a paper by reading the abstract carefully and noting the four kinds of informationoutlined above. Then move first to the visuals and then to the rest of the paper.

Features of Introductions

Introductions serve two purposes: creating readers' interest in the subject and providingthem with enough information to understand the article. Generally, introductions accomplish this by leading readers from broad information (what is *known* about the topic) to more specific information (what is *not known*) to a focal point (what *question* the authors asked and answered). Thus, authors describe previous work that led to current understanding of the topic (the broad) and then situate their work (the specific) within the field.

Features of Methods

The Methods section tells the reader what experiments were done to answer the questionstated in the Introduction. Methods are often difficult to read, especially for graduate students, because of technical language and a level of detail sufficient for another trained scientist to repeat the experiments. However, you can more fully understand the design of the experiments and evaluate their validity by reading the Methods section carefully.

Features of Results and Discussion

The Results section contains results—statements of what was found, and reference to thedata shown in visuals (figures and tables). Normally, authors do not include informationthat would need to be referenced, such as comparison to others' results. Instead, that material is placed in the Discussion—placing the work in context of the broader field.

The Discussion also functions to provide a clear answer to the question posed in theIntroduction and to explain how the results support that conclusion.

Atypical Structure

Some articles you read will deviate from the conventional content of IMRD sections. Forinstance, Letters to *Nature* appear to begin with an abstract, followed by the body of the article. Upon reading, however, you will see that the "abstract" is a summary of the workfilled with extensive introduction (for the purpose of catching the attention of a wide audience), and the next paragraph begins a description of the experiments.

Therefore, when you begin to read an article for the first time, skim the article to analyzethe document as a whole. Are the sections labeled with headings that identify the structure? If not, note what the structure is. Decide which sections contain the material most essential to your understanding of the article. Then decide the order in which you will read the sections.

2. Distinguish main points.

Because articles contain so much information, it may be difficult to distinguish the *main points* of an article from the *subordinate points*. Fortunately, there are many indicators ofthe author's main points:

Document level

- Title
- Abstract
- Keywords
- visuals (especially figure and table titles)
- first sentence or the last 1-2 sentences of the Introduction

Paragraph level: words or phrases to look for

- *surprising*
- *unexpected*
- *in contrast with previous work*
- *has seldom been addressed*

- *we hypothesize that*
- *we propose*
- *we introduce*
- *we develop*
- *the data suggest*

3. Generate questions and be aware of your understanding

Reading is an active task. Before and during your reading, ask yourself these questions:

- Who are these authors? What journal is this? Might I question the credibility of the work?
- Have I taken the time to understand all the terminology?
- Have I gone back to read an article or review that would help me understand this work better?
- Am I spending too much time reading the less important parts of this article?
- Is there someone I can talk to about confusing parts of this article?

After reading, ask yourself these questions:

- What specific problem does this research address? Why is it important?
- Is the method used a good one? The best one?
- What are the specific findings? Am I able to summarize them in one or two sentences?
- Are the findings supported by persuasive evidence?
- Is there an alternative interpretation of the data that the author did not address?
- How are the findings unique/new/unusual or supportive of other work in the field?
- How do these results relate to the work I'm interested in? To other work I've read about?
- What are some of the specific applications of the ideas presented here? What are some further experiments that would answer remaining questions?

4. Draw inferences.

Not everything that you learn from an article is stated explicitly. As you read, rely on your prior knowledge and world experience, as well as the background provided in thearticle, to draw inferences from the material. Research has shown that readers who actively draw inferences are better able to understand and recall information.

As an example, in the box below is an excerpt from the Introduction of an article in the journal *Biochemistry (Ballestar et al., 2000)*. The comments in italics are questions andinferences that might be drawn by a student reader.

Rett Syndrome is a childhood neurodevelopmental disorder and one of the most common causes of mental retardation in females *Comment: Hmmm…must be related to a gene on the X-chromosome*, with an incidence of 1 in 10000-15000. *Comment: How common is that? Not too likely to happen to me, but there must be several such children born in Houston every year.* Rett syndrome patients are characterized by a period of normal growth and development (6-18 months) followed by regression with loss of speech and purposeful hand use. *Comment: What happens? Something must be triggered or activated at late infancy.* Patients also develop seizures, autism, and ataxia. After initial

regression, the condition stabilizes and patients survive into adulthood. Studies of familial cases provided evidence that Rett is caused by X-linked dominant mutations in a gene subject to X-chromosome inactivation. Recently, a number of mutations in the gene encoding the methyl-CpG binding transcriptional repressor MeCP2 have been associated with Rett Syndrome. *Comment: MeCP2 mutations probably cause Rett Syndrome. This must be an important master-regulator to affect so many processes in the brain. I wonder what they know about it…*

5. Take notes as you read.

Effective readers take notes—it improves recall and comprehension. You may think you'll remember everything you read in researching class assignments, professional papers, proposals, or your thesis, but details will slip away. Develop a template for recording notes on articles you read, or adapt the template below for use. As you accumulate a large collection of articles, this template will help you distinguish articles and quickly locate the correct reference for your own writing. The time spent filling outthe form will save you hours of rereading when you write a Background, Related Work,or a Literature Review section.

Template for Taking Notes on Research Articles:
Easy access for later use

Whenever you read an article, pertinent book chapter, or research on the web, use the following format (or something similar) to make an electronic record of your notes for later easy access. Put quotation marks around any exact wording you write down so that you can avoid accidental plagiarism when you later cite the article.

Complete citation. Author(s), Date of publication, Title (book or article), Journal, Volume #, Issue #, pages:

If web access: url; date accessed

Key Words:

General subject:

Specific subject:

Hypothesis:

Methodology:

Result(s):

Summary of key points:

Context (how this article relates to other work in the field; how it ties in with key issues and findings by others, including yourself):

Significance (to the field; in relation to your own work):

Important Figures and/or Tables (brief description; page number):

Cited References to follow up on (cite those obviously related to your topic AND any papers frequently cited by others because those works may well prove to be essential as you develop your own work):

Other Comments:

References

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Reading a Computer Science Research Paper

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Abstract

This tutorial article highlights some points that a graduate or senior undergrad-uate student should bear in mind when reading a computer science research paper. Specifically, the reading process is divided into three tasks: comprehension, evaluation and synthesis. The genre of paper review is then introduced as a vehicle for critical reading of research papers. Lastly, guidelines on how to be initiated into the trade of conference and/or journal paper review are given.

Designed to be used in a graduate course setting, this tutorial comes with a sug- gested marking scheme for grading paper reviews with a comprehension-evaluation- synthesis structure.

1 Comprehension

The first lesson to reading research papers is learning to understand what a paper says. A common pitfall for a beginner is to focus solely on the technicalities. Yes, technical contents are very important, but they are in no way the only focus of a careful reading. In general, you should ask yourself the following four questions when you are reading a research paper.

1. What is the research problem the paper attempts to address? What is the *motivation* of the research work? Is there a *crisis* in the research field that the paper attempts to resolve? Is the research work attempting to overcome the *weaknesses* of existing approaches? Is an existing *research paradigm* challenged? In short, what is the *niche* of the paper?

2. What are the claimed contributions of the paper? What is *new* in this paper? A new *question* is asked? A new *understanding* of the research problem? A new *methodology* for solving problems? A new *algorithm*? A new breed of software *tools* or

systems? A new *experimental method* ? A new *proof technique*? A new *formalism* or *notation*? A new *evidence* to substantiate or disprove a previously published claim? A new *research area*? In short, what is *innovative* about this paper?

3. How do the authors substantiate their claims? What is the *methodology* adoptedto substantiate the claims? What is the *argument* of the paper? What are the major *theorems*? What *experiments* are conducted? *Data analyses*? *Simulations*? *Bench- marks*? *User studies*? *Case studies*? *Examples*? In short, what makes the claims *scientific* (as opposed to being mere opinions[1])?

4. What are the conclusions? What have we *learned* from the paper? Shall the *standard practice* of the field be changed as a result of the new findings? Is the result *generalizable*? Can the result be applied to *other areas* of the field? What are the *open problems*? In short, what are the *lessons* one can learn from the paper?

Every well-written research paper contains an *abstract*, which is a summary of the paper.The role of an abstract is to outline the answers to the above questions. Look therefore, firstto the abstract for answers. The paper should be an elaboration of the abstract.

Another way of looking at paper reading is that every good paper tells a *story*. Conse- quently, when you read a paper, ask yourself, "What is the plot?" The four questions listed above make up an archetypical plot structure for every research paper.

2 Evaluation

An integral component of scholarship is to be critical of scientific claims. Ambitious claims are usually easy to make but difficult to substantiate. Solid scholarship involves careful validation of scientific claims. Reading research papers is therefore an exercise of critical thinking.

1. Is the research problem significant? Is the work scratching *minor itches*? Are the authors solving *artificial problems* (aka *straw man*)? Does the work enable *practical applications*, deepen *understanding*, or explore *new design space*?

2. Are the contributions significant? Is the paper *worth reading* ? Are the authors simply *repeating* the state of the art? Are there real *surprises*? Are the authors aware of the relation of their work to *existing literature*[2]? Is the paper addressing a well-known*open problem*?

[1]Alternatively, what makes it a research paper rather than a *science fiction*?

[2]Be very sceptical of work that is so "*novel*" that it bears no relation to any existing work, builds upon no existing paradigm, and yet addresses a research problem so significant that it promises to transform the world. Such are the signs that the author might not be aware of existing literature on the topic. In such a case, the authors could very well be repeating works that have already been done decades ago

3. Are the claims valid? Have the authors been *cutting corners* (intentionally or un-intentionally)? Has the right theorem been proven? Errors in proofs? Problematic experimental setup? Confounding factors? Unrealistic, artificial benchmarks? Com- paring apples and oranges? Methodological misunderstanding? Do the numbers add up? Are generalizations valid? Are the claims modest enough[3]?

When you evaluate a research work, two caveats are worth noting:

- Consistently evaluating research works in a negative way gives a young researchera false sense of being critical. Learn to be fair: attend to both the strengths and weaknesses of the work. If you are reading a classical paper that has been published for a while, make sure you are reading the paper in the right historical context: What seems to be obvious now might have been ground-breaking then.
- A young researcher may want to focus on point 3 (Are the claims valid?). Evaluating the significance of the research problem and the contributions of the paper usually requires a comprehensive understanding of the research field as a whole. Yet, do not let the above comment hinder you from disagreeing with the paper authors in mattersof significance.

3 Synthesis

Creativity does not arise from the void. Interacting with the scholarly community through reading reseach papers is one of the most effective ways for generating novel research agendas. When you read a research paper, you should see it as an opportunity for you to come up with new research projects. The following is a list of questions you can ask to help in this direction. (Of course, this list is not supposed to be exhaustive.)

- What is the crux of the research problem?
- What are some alternative approaches to address the research problem?
- Is there an alternative way to substantiate the claim of the authors?
- What is a good argument against the case made by the authors?
- Can the research results be strengthened?
- Can the research results be applied to another context?
- What are the open problems raised by this work?
- *Bottom line:* If you were to do the research, how would you do differently?

[3]It is very tempting for an inexperienced researcher to make overly general conclusion from limited evidence. A high quality scientific claim is always *modest* — claiming only what can be concluded from the evidence, making explicit the limitation of the evidence, and carefully delimiting the scope of the claim.

4 Paper Review

A paper review is a short essay (5 pages, single-space) reporting what you have learned from reading a research paper. Writing reviews for the papers you have read is a great way to sharpen your paper reading skills. Such a review is typically structured in three sections — *summary*, *critique*, and *synthesis*.

1. Summary. Give a brief summary of the work *in your own words*. This section demonstrates your understanding of the paper, and as such it should answer the four questions outlined in Section 1. The summary section should be structured as follows: (1) motivation, (2) contribution, (3) methodology and/or argument, and (4) conclusion.

   It is imperative that you use your own words to summarize the paper. Failing to adhere to this guideline not only constitutes plagarism, but also demonstrates that you probably do not quite understand the work. You can be sure that you understand something only when you are capable of explaining it in your own words.

2. Critique. Pick two to three points[4] you want to argue with the authors[5]. Use the questions outlined in Section 2 to help you come up with meaningful critiques.

   Do not repeat the Limitations section of the paper. Doing so means that you *agree* with the authors! Pick points of *disagreement*, and launch an intellectual debate with the authors. Carefully articulate and substantiate your case. Do not just say, "I don't like this point." Instead, give technical reasons to substantiate your critiques.

   Be specific in your choice of words. Avoid generic adjectives such as "bad", "poor", "lame", "stupid", etc, and their synonyms and antonyms. You can go a lot fur-ther by replacing such vague words with more specific ones: "inelegant", "inefficient", "memory-intensive", "ill-defined", etc.

3. Synthesis. Propose one to two ways in which the research work can be further developed. Do not repeat the *Future Work* section of the paper. Be original. Consult the list of questions in Section 3 if you run out of ideas.

I use this format when I ask students in my graduate classes to review a paper. Consult the Appendix for a sample outline, a marking scheme, and page length suggestions.

[4]This restriction on the number of critiques is intentional. Firstly, the restriction forces you to pinpoint the major weaknesses of the paper, rather than to spend efforts debating issues of peripheral importance. Secondly, such a restriction allows you to enjoy the mental room necessary for developing a substantial case against the authors.

[5]Notice that the Critique section presents only negative evaluations of the paper. Have we forgotten about being fair to a research work? No, positive evaluations are omitted for a good reason. Experience tells us that students tend to give positive evaluation in the following form: "I agree with the authors. They did this and that, and they did a good job." The end result is usually a repetition of the authors' claims. Ifind that focusing on critiques offers a more substantial learning experience to the students, forcing them to think rather than to parrot.

5 An Alternative Review Format

The format of paper review outlined in the previous section is the one I adopt for my graduate classes. It works for me, but it is definitely not the only way to structure a paper review. I outline here an alternative format I learned from a friend of mine.

1. *What is the purpose of the work?*
2. *How do the authors achieve this purpose? Why is this particular approach adopted?*
3. *Do you think the purpose has been achieved?*
4. *What insights have you gained from reading this work?*

Notice the parallel between this alternative structure and the summary-critique-synthesis structure in the previous section.

6 Reading Like a Pro

When a research paper is submitted to a conference or a journal, it will undergo a *peer review* process, in which the paper is subject to the intense scrutiny of peer researchers. The *referees* who review the submitted paper will read the paper in more or less the same way as we outlined in Sections 1 and 2, and then they will write up a *referee report* in a style similar to the paper review discussed in Section 4, except for the synthesis section. Based on the referee reports, the program chair of a conference or the editor of a journal will then make the decision of whether to accept the paper. It is therefore instructional to understand how a referee go about reviewing a paper, and learn to read research papers like a professional. A very good introduction to the subject can be found in an article by Smith [2]. The paper is slanted towards experimental computer science. For a perspective focusing on theoretical computer science, consult the article by Parberry [1]. After reading these papers, I highly recommend graduate students to find opportunities to practice professional paper reviewing. Your supervisor will likely be involved in the program committees of conferences, or asked to review papers for conferences. Approach your thesis supervisors, and volunteer to help out with paper reviews. By actually writing up a professional review report, and discussing your review with your supervisor, you will gain tremendous insight into the paper publishing process, as well as the implicit value system of the academic world. This kind of training is hard to acquire through other means.

References

[1] Ian Parberry. A guide for new referees in theoretical computer science. *Information and Computation*, 112(1):96–116, July 1994.

[2] Alan Jay Smith. The task of the referee. *IEEE Computer*, 23(4):65–71, April 1990.

A Sample Outline of a Paper Review

Review: *Title of Paper being Reviewed*

Author of this Review

Other information such as student ID

1 Summary (40% : 2.5 pages)

1.1 Motivation (8%)

1.2 Contribution (8%)

1.3 Methodology (16%)

1.4 Conclusion (8%)

2 Critique (30% : 1.5 pages)

2.1 Title of 1st Critique (15%)

2.2 Title of 2nd Critique (15%)

2.3 Optional: Title of 3rd Critique

If a 3rd critique is given, then each critique is worth 10%.

3 Synthesis (30% : 1 page)

3.1 Title of 1st Idea (30%)

3.2 Optional: Title of 2nd Idea

If a 2nd idea is presented, then each idea is worth 15%.

Researcher Bias in Software Engineering Experiments: a Qualitative Investigation

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***Abstract*—Researcher Bias (RB) occurs when researchers influence the results of an empirical study based on their expectations. RB might be due to the use of Questionable Research Practices (QRPs). In research fields like medicine, blinding techniques have been applied to counteract RB. We conducted an explorative qualitative survey to investigate RB in Software Engineering (SE) experiments, with respect to: *(i)* QRPs potentially leading to RB, *(ii)* causes behind RB, and *(iii)* possible actions to counteractRB including blinding techniques. Data collection was based on semi-structured interviews. We interviewed nine active experts inthe empirical SE community. We then analyzed the transcripts of these interviews through thematic analysis. We found thatsome QRPs are acceptable in certain cases. Also, it appears that the presence of RB is perceived in SE and, to counteract RB, a number of solutions have been highlighted: some are intended for SE researchers and others for the boards of SE research outlets.**

***Index Terms*—Survey, interview, researcher bias, blinding.**

I. Introduction

In research, *bias* is defined as the combination of various design, data, analysis, and presentation factors tending to produce findings that should not be produced [1]. *Researcher Bias* (*RB*), or *experimenter bias*, occurs when researchers, consciously or unconsciously, influence the results of an empirical study based on their expectations. In some cases, RB is due to the adoption of Questionable Research Prac- tices (QRPs) to follow one's agenda and achieve specific expectations—*e.g.,* adjusting an experimental sample until statistically significant results are found. Another form of bias is *publication bias*, which occurs when studies are published based on their results—usually positive results are more likely to be published than negative ones [2]. To counteract RB, according to established guidelines in Software Engineering (SE), researchers should disclaim their stance regarding an outcome. For example, Wohlin *et al.* [3] consider *experimenter expectancies* as a social threat to construct validity. Nevertheless, it has been shown that RB affects SE studies [4], [5].

In this paper, we report the opinions of a group of experts about themes related to RB in SE experiments. To this end, we conducted a qualitative explorative survey. The scope of our research is both on human- and technology- oriented experiments [3]. We are concerned with: QRPs related toRB, causes behind RB, and possible actions to counteractit. Regarding the latter, we focus on two techniques—*blind data extraction* and *blind data analysis*. The former consistsof hiding some information (*e.g.,* treatment assignment) from the researchers who extract the data; whereas, the latter isthe temporary and judicious removal of labels and alterationof values before someone analyzes the data [6]. Although extensively used in other research fields like medicine and physics [6], [7], SE researchers have used these techniques in few occasions [8], [9]. To collect data, we conducted semi-structured interviews. Nine experts in the field of empirical SE took part in these interviews. We then applied thematic analysis [10] to organize experts' opinions.

II. Background

In this section, we survey the status of RB in SE, along with QRPs associated to it. Moreover, we describe countermeasures to RB including blinding techniques, which our survey explicitly focuses on.

A. Questionable Research Practices and Researcher Bias

Cases of QRPs, exploiting the gray area of what is considered acceptable, have been mounting in medicine, natural sciences, and psychology (*e.g.,* [11], [12]). As for the SE research field, Jørgensen *et al.* [5] documented the presence of RB and publication bias in SE experiments. The authors conducted a quantitative questionnaire-based survey, with researchers from some SE sub-communities, comprising questions about QRPs potentially leading to RB and publication bias. Three out of seven questions were on QRPs related to RB, namely:

1) *Post-hoc hypotheses*—defined as reporting the results of one (or more) hypothesis tests where at least one of the hypotheses is formulated after looking at the data.
2) *Post-hoc outlier criteria*—defined as developing or changing the rules for excluding data (*e.g.,* outlier removal) after looking at the impact of doing so on the results.

3) *Flexible reporting of measures and analysis*—defined as using several variants of a measure or several tests and then reporting only the measures and tests that give the strongest results.

The results suggest that: *(i)* 67% of the respondents had followed the post-hoc hypotheses practice; *(ii)* 55% had followed the post-hoc outlier criteria practice; and *(iii)* 69% had followed the flexible reporting of measures and analysis practices. Jørgensen *et al.* [5] also built a model—150 randomly-sampled SE experiments fed the model—to estimate the proportion of correct results at different levels of RB and publication bias. The model suggests that both RB and publication bias affect SE experiments since 52% of the statistically significant tests do not match a situation with no or low RB and publica-tion bias. Shepperd *et al.* [4] in their meta-analysis of defect prediction techniques came to a conclusion similar to that by Jørgensen *et al.* [5]. The authors pointed out the presence of RB in the studies included in the meta-analysis as the factor with the largest effect was the research group publishing the paper, while the effect of the prediction technique was small.

B. Countermeasures to Researchers Bias

Potential solutions to counteract RB have been pro-posed (*e.g.,* [13], [14]). We can group them into: *(i) rival theories*; *(ii) transparency*; and *(iii) blinding*. The first categoryconsists of considering alternative or competing hypotheses with respect to the ones being tested in the study. The researcher should then devise experiments that can explicitly distinguish competing hypotheses and, if possible, develop experiments that can distinguish between alternative theories. The researcher should collaborate with a *team of rivals—i.e.,* other researchers that, while being skeptical about the hypothe- ses, collaborate towards developing alternative explanations.

Several approaches fall under the umbrella of the transparency category. The prime example is *open science*, *i.e.,* the practice of sharing research data, computer code, and lab packages for public scrutiny so attempting to reproduce results. In fields like medicine or psychology, transparency is also achieved through *pre-registration* (or *registered report*).It consists of submitting a paper presenting the study rationale and planning for peer review before its conduction. Once the paper is accepted, the researchers can conduct the study and submit a paper with the obtained results for a second round of revision. The paper cannot be rejected due to the study results (*e.g.,* negative results) while it can be for other reasons (*e.g.,* deviations from the pre-registered analysis procedure) [15].

Finally, blinding (or *masking*) means concealing research design elements (*e.g.,* treatment assignment or research hypotheses) from individuals involved in an empirical study (*e.g.,* participants, data collectors, or data analysts) [16], [17]. The use of blinding techniques has been encouraged fields like medicine and physics [6], [7]. As for SE, Shepperd *et al.* [4] have fostered researchers to use blinding techniques in their studies; however, few researchers have applied blinding in SE studies so far, namely: Fucci *et al.* [8], who used blind data extraction and analysis in a human-oriented experiment; and Sigweni and Shepperd [9], who applied blind data analysis in a technology-oriented experiment.

To explain how blind data extraction and analysis work, let us take as an example the experiment by Fucci *et al.* [8]. The study goal was to investigate the claimed effects of Test-Driven Development (TDD), such as the increase in developers' productivity. The experiment compared a *treatment group—i.e.,* developers who applied TDD to implement some programs—to a *control group—i.e.,* developers who implemented thesame programs as the other group but by applying Test-Last Development (TLD). After the experiment took place, the raw dataset (*i.e.,* the programs the developers implemented) was given to a researcher that played the role of data extractor. Given the raw dataset, this researcher extracted the values of the metrics (*e.g.,* the PROD metric quantified developers' productivity) so obtaining the dataset. The extraction of the metrics was done blindly because the data extractor was not aware of the experimental goal, hypotheses, treatment assignment, and design. The dataset was then given to the data analysts who performed the blind data analysis. They were not aware of the experimental goal and they worked on a sanitized dataset—*i.e.,* the values of the independent variables were temporarily replaced (*e.g.,* the TDD and TLD groups became the A and B groups, respectively) and the dependent variables were temporarily anonymized (*e.g.,* PROD was renamed as DV1). To correctly analyze the data, the analysts were provided with a minimal description of the dependent and independent variables (*e.g.,* DV1 assumed values in $[0, 1]$) as well as the experimental design in which some information was adequately hidden (*e.g.,* the experimental groups were referred as A and B). One the analysis was complete, the hidden information was revealed.

III. Study Design and Limitations

A. Goal

Unlike quantitative research that seeks to provide quantifiable responses to some research questions, qualitative research (like ours) is concerned with understanding subjects' viewpoints about a given phenomenon and discovering the causes behind that phenomenon as noticed by the subjects [3]. Therefore, the goal of our study is to elicit the opinions of a group of experts about RB in SE experiments, including: *(i)* QRPs potentially leading to RB; *(ii)* causes behind RB; and *(iii)* potential actions to counteract RB with a focus on, but not limited to, blind data extraction and analysis. That is to say that, while the studies by Shepperd *et al.* [4] and Jørgensen *et al.* [5] show the presence of RB in SE studies by leveraging quantifiable responses, we are more interested in investigating the phenomenon of RB based on researchers' viewpoints. Our study can be thus considered complementary to those above-mentioned [4], [5].

B. Protocol

We planned a series of interviews with experts from the empirical SE community to investigate on RB. Despite interviewees are time-consuming, we opted for this data collec-

TABLE I: Characterization of the interviewees involved in this study.

ID	Institution region	Academic position	Main research interest	Experience as experimenters	Last published experiment
R1	Southeastern Europe	Assistant professor	Defect prediction	5-10 (years)	< 6 months
R2	Northern Europe	PhD student	Human and social aspects of SE	1-5 (years)	< 18 months
R3	Northern Europe	Full professor	Mining software repositories	11-20 (years)	< 6 months
R4	Northern America	Associate professor	Agile software development	11-20 (years)	< 6 months
R5	Central Europe	Assistant professor	Software maintenance and evolution	5-10 (years)	< 3 years
R6	Southern Europe	Associate professor	Software economics and metrics	11-20 (years)	< 1 year
R7	Southern Europe	Assistant professor	Project and process management	11-20 (years)	< 1 year
R8	Southern Europe	Full professor	Collaborative software development	> 20 (years)	< 18 months
R9	Southern Europe	Full professor	Software economics and metrics	11-20 (years)	< 6 months

tion means, rather than questionnaires, because: *(i)* it allows achieving higher response rates; *(ii)* it decreases the number of "don't know" and "no answers"; and *(iii)* the interviewer can ask for clarifications if needed [3]. Also, such a methodfits the explorative intention of our study.

We recruited researchers in our network based on their experience in SE experimentation (both human- and technology-oriented). Nine researchers were available for an interview either face-to-face or by phone. Each interview involved the same interviewer (*i.e.,* DF) together with one interviewee at a time. We obtained consent from the interviewee to be audio-recorded and we informed each of them that the gathered data would be treated confidentially. Each interview lasted between 50 and 75 minutes. We used semi-structured interviews [3]. That is, the questions listed in the interview script were not necessarily asked in order because, depending on how the conversation evolved, some questions were handled beforethan others. Semi-structured interviews enable improvisation and exploration of the investigated phenomenon. The interview script serves to guide the discussion and make sure that relevant topics are covered [3].

The interview script[1] consisted of eight parts. The objective of the first part (*i.e., Warm Up*) was to gather demographic information on the interviewees (*e.g.,* where the researcher was employed or her research interests). This information allowed us to characterize the study context. As for the second part (*i.e., Experiments*), the interviewer asked to guide him throughthe usual experimental process of the interviewee. The goal was to break the ice between interviewer and researcher by gathering information on how researchers design an experi- ment and the division of work in case a team of researchersis involved. We were also interested in their perception of threats to validity that can arise given their design choices.In the third part (*i.e., QRP*), we gathered the interviewees' viewpoints on some QRPs recently reported in the survey by Jørgensen *et al.* [5]. We considered those QRPs potentially leading to RB, namely: post-hoc hypotheses, post-hoc outlier criteria, and flexible reporting of measures and analyses. We did not consider QRPs related to publication bias because our paper does not focus on this kind of bias. In the fourth part (*i.e., RB*), we focused on how the interviewees perceived RB in SE experiments, the causes behind it, and their suggestions to avoid/mitigate it. With the fifth (*i.e., Blind Data Extraction*) and sixth (*i.e., Blind Data Analysis*) parts, we centered the discussion on blind data extraction and analysis, respectively. We gathered opinions about the aforementioned techniques to cope with RB. As for the seventh part (*i.e., Blind Data Extraction and Analysis*), we focused on the interviewees' thoughts on the use of blind data extraction and analysis together, as well as how to foster the use of these techniques. We ended each interview (*i.e., Wrap Up*) by asking whether the interviewee would use blind data extraction and analysisin her future experiments.

C. Participants

All the interviewees had, at the time of our study, published at least one experiment in one of the SE higher quality venues (*i.e.,* ICSE, EMSE, IEEE TSE, and ACM TOSEM). In Table I, we report some interviewees' information gathered in the *Warm Up* part of the study. We guarantee the anonymity of the interviewees by referring to each of them through an ID (from R1 to R9). As Table I shows, the participates were quite heterogeneous in terms of location of their institution, academic position, main research interest, years of experience as experimenter, and date of last published experiment.

D. Data Analysis

After transcribing the recordings of the interviews, we (*i.e.,* SR, MTB, and GS) analyzed the transcripts using a thematic analysis approach called template analysis, which is known to be flexible and fast [10]. Template analysis allows developing a list of codes, each of which identifies a theme within the transcripts. The codes are arranged in a *template*—it usually is a hierarchical structure of codes—showing the relationships among themes, as defined by the investigators. In template analysis, the investigators start analyzing the transcripts by using an initial template. That is, they attach pre-defined codes, arranged in a template, to delimit portions of text related to themes. As King [10] suggests, the best starting point for developing an initial template is the interview script. Accordingly, we developed our initial hierarchical template (see the non-bold text in Figure 1) from the interview script. As customary in template analysis, we revised the initial template during the analysis [10]. In particular, we renamed the second-level code *Presence of Researcher Bias* as *Presence of Researcher Bias and Clues* because we found portions of text about clues suggesting the presence of RB. We concludedthe analysis when any portion of text relevant to the goal of our study was coded and we agreed on the obtained template. To ease the thematic analysis, we used the *ATLAS.ti* tool.

E. Limitations

When interpreting the findings from qualitative investigations, some limitations have to be taken into account:

- Experiments
  - Researcher Roles
  - Threats to Validity
- Questionable Research Practices
  - Post-hoc Hypotheses
  - Post-Hoc Outlier Criteria
  - Flexible Reporting of Measures and Analyses
- Researcher Bias
  - Presence of Researcher Bias **And Clues**
  - Causes of Researcher Bias
  - Coping with Researcher Bias
    - Blind Data Extraction
      - Usefulness of Blind Data Extraction
      - Drawbacks of Blind Data Extraction
    - Blind Data Analysis
      - Usefulness of Blind Data Analysis
      - Drawbacks of Blind Data Analysis
    - Blind Data Extraction and Analysis
      - Effectiveness of Blind Data Extraction and Analysis
      - Fostering Blind Data Extraction and Analysis

Fig. 1: Initial and final templates (in bold, text added to the initial template to obtain the final one).

- The interviewees may not answer truthfully because, for example, they are scarcely motivated or afraid of being judged. To mitigate this threat, the participation in the study was voluntary—volunteers are generally more motivated [3]—and we informed the interviewees about data confidentiality.
- The number of the interviews might threaten the validity of results. However, Guest *et al.* [18] observed that a sample of six interviews may be sufficient to allow development of meaningful themes and useful interpretations. Given the results observed after the analysis of the nine interviews, we believe to have hit a point of diminishing return [18] for which increasing the number of interviews will unlikely generate more evidence. Moreover, our plan includes (quantitative) surveys, based on questionnaires, with researchers from different SE sub-communities (*e.g.,* ICSE or ESEM) to understand how much they agree with the interviewees' statements (*i.e.,* we are going to apply *methodological triangulation* [19]) and whether or not there are differences among different SE sub-communities.
- Our findings might not generalize to researches sampled from a different population. As previously mentioned, we are going to investigate this point in our long-term planwith quantitative surveys.
- The investigator might unconsciously influence the results based on its expectations. We mitigated such a threat by involving more people in conducting and analyzing the interviews. In particular, DF was the interviewer while SR, MTB, and GS performed the data analysis (*i.e.,* we applied *investigation triangulation* [19]).

IV. Findings

We present the findings emerging from the interviews based on the main themes identified by the first-level codes (*e.g., Questionable Research Practices*) of the final template shown in Figure 1. To bring credibility to our findings, we present them together with some excerpts from the transcripts.

A. Experiments

As shown in Figure 1, two sub-themes were defined within this main theme: the roles of researchers in SE experiments and how they cope with threats to validity in their experiments.

Researcher Roles. It emerged that, when conducting an experiment, there is a division of roles among the researchers. Each researcher covers one or more roles (*e.g.,* one researcher is involved in the planning of the experiment and in its execution, another one extracts the metrics from the raw data and so on). However, it seems that only one researcher takes care of data analysis (*i.e.,* one researcher plays the data analyst role). An excerpt from the interview with R6 follows:

> We [our research group] outlined the experiment design. The researchers from [other country] translated the experiment material into [other language] and carried out the experiment in [other country]. We then received the gathered data, some Excel files, and one of us executed the analysis.

Threats to Validity. When we asked the interviewees to elaborate on threats to validity, they provided a number of examples, but none mentioned RB.

B. Questionable Research Practices

This theme includes three sub-themes (see Figure 1): the participants' perceptions of post-hoc hypotheses, post-hoc outlier criteria, and flexible reporting of measures and analyses.

Post-hoc Hypotheses. According to the interviewees, post-hoc hypotheses should not to lead to RB as long as: *(i)* the researchers clearly report that such hypotheses are formulated in retrospect; or *(ii)* it is possible to ground such hypotheses on prior work (thus, there is no need to make clear that such hypotheses are post-hoc). Regarding *(i)*, R5 said:

> In this case, first of all I am not sure we can talk about formulating hypotheses because you are already looking at the data of an experiment [...] In general, I don't think there is anything wrong with that if, and Ithink it is completely sound, if you explicitly say that it is an unexpected result when reporting this result. This is different from saying "we wantedto investigate this and we found that it is supported by the data."

As for the point *(ii)*, R3 said:

> Of course, there's the fact that, the hypothesis should be grounded on prior work. If you can ground something to solid prior work, then it doesn't really matter whether it was sort of after the fact.

Also, it seems post-hoc hypotheses could be a means to get new insight into the studied phenomenon, which researchers had not thought about when the study was planned. R4 said:

> It [a post-hoc hypothesis] emerged from the data and inevitably happens. When you look at the data, you may have, you may think of new insights that you haven't thought about because there is information that was not anticipated. [...] Sometimes there are research methodologies that don't even assume any questions, they are completely totally exploratory. So let's suppose that you have a set of questions, and you wanna answer them first. After you answer those questions, then you see some other patterns in your data and then, in the next iteration, you formulate a set of other questionsthat maybe you can answer based on the same data. This is completely okay but it's not the same as fishing.

Post-hoc Outlier Criteria. The interviewees seem to believe that this practice should be avoided because it potentially leads to RB, though not necessarily. R5 told us:

> Looking at the results and then removing outliers could sometimes be sensible, but I think the bias would be too strong.

In case researchers apply the post-hoc outlier criteria practice, the interviewees agreed that they should declare the use of this practice in the paper by providing, for example, the following information: *(i)* the results before and after removing outliers; *(ii)* the reasons behind the outlier removal; and *(iii)* an interpretation of the results (*e.g.,* why, after the outlier removal, a null hypothesis passes from non-rejected to rejected). R4 said:

> As long as you declare the results and you present maybe both of them [before and after the outlier removal], depending on how other factors influence your interpretation. Maybe there are other things that you discovered during your data analysis that justifies that decision. But as long as you declare them, I mean that is one of the purposes of the peer review, the reviewers can also decide which one is, whether that decision was sensible or not.

Flexible Reporting of Measures and Analysis. Based on interviewees' experience, when researches can choose among equivalent statistical hypothesis tests (*e.g.,* t-test or F-test), the results (*i.e.,* p-values) are not so different. R8 told us:

> It's true that there are a lot of statistical hypothesis tests and there are a lot of variants as well, when using statistical packages we are spoilt for choice, but in my experience they don't vary so much.

Furthermore, according to R3, if a statistical hypothesis test revealed a significant difference that an equivalent test did that difference would be probably negligible. In other words, the effect size would show the true impact of that difference, so having or not a significant difference would not matter:

> It [using a statistical hypothesis test or an equivalent one] doesn't really impact the results very much. It's a very very tiny difference, at least whatI have seen. It doesn't change from .04 to .0004, or something. I mean you might, if you again use this magical threshold of .05, then it might matter. But if you report the effect sizes, then it really doesn't. The effect sizes sortof reveal the true impact.

As for the practice of using several variants of a measure and then reporting only the variants that give the strongest results, it is perceived as a bad practice. The researchers should discuss any variant of that measure in the paper. R4 said:

> Yeah I think that is a no, in general. If you've done [flexible reporting of measures], there needs to be a discussion of how your attempt to triangulate the results with different measures failed. That should be partof the discussion and it's part of the validity threats that you have.

C. Researcher Bias

This theme has three sub-themes (see Figure 1): the presence of RB in experiments and clues suggesting such a presence; causes of RB; and strategies to cope with RB.

Presence of Researcher Bias and Clues. From the interviews, it emerged that RB affects the SE community. Although the interviewees did not have proofs about the presence of RB in SE, they pointed out four clues suggesting its presence: *(i)* RB affects any community (*e.g.,* medicine or psychology); *(ii)* when reviewing papers, it is not rare to suspect authors biasing the results; *(iii)* whoever could unconsciously biasthe results based on her expectations; and *(iv)* there are sometimes inconsistent results among studies investigating the same constructs. On the points *(i)* and *(ii)*, R4 stated:

> I think it [RB] must be happening because it's probably happening in every community. But I'm not sure. I mean I think, in terms of my review work, when things are suspicious, it's usually obvious and it's usually not just from one reviewer picking on them, rather, multiple reviewers do and it's only because, the researchers actually let it be understood in the paper.

As for the point *(iii)*, R3's thought follows:

> I guess everyone that does experiments is somehow biased because you know that negative results cannot be published and it probably, sort of unconsciously, alters your actions.

On the last point, R8 said:

> That is, if I see that a given result isn't confirmed [by another study], then it is a clue of researcher bias.

Causes of Researcher Bias. Four causes of RB emerged from the interviews. First, interviewees believed that *negative-results papers are usually rejected*. This would lead re- searchers to bias their results (*e.g.,* transforming non- significant results into statistically significant ones). R2 said:

> I think the main reason to that is there is no acceptance for reporting the negative results. You are a researcher and your responsibility is just to explore the phenomenon, whether it is in favor of your hypothesis or it's against your hypothesis you should report it, but I've personally felt like there is no in general acceptance for that.

Second, the *pressure of publishing papers* can lead researchers to (unconsciously or consciously) bias the results. R5 said:

> Especially young researchers, for example Ph.D. students, that carry out and are therefore responsible for the experiment, may tend to have high expectations on what they have developed or towards the hypothesis being verified, to the point that, even unconsciously, they may tend to guide the experiment towards a certain expected result. I am quite confident to say that, although not always, this occurs especially with novice experimenters that are more eager for publications and may therefore be led to experimenter bias.

Third, it seems that *revision processes of SE conferences/journals are focusing too much on the empirical assessment*, rather than on the contributions of the ideas to the body of knowledge. Thus, researchers would be led to bias their studies by making the results more publishable. R5 told us:

> I think that the main problem of several review processes is that they are highly based on the empirical aspect and much less on the novelty of the ideas. So in spite of you propose an interesting and novel idea that several other researchers can build on, if the experimental results are not strong enough you are likely to receive a comment like "okay nice idea but ...".On the other hand, if a study is empirically perfect, from the point of viewof the design and results, but has very limited novelty, it's difficult that itwill be rejected.

Fourth, the *immaturity of the SE field and its researchers*. That is, some researchers believe not to bias the results of their experiments when they actually do. In this respect, R9 said:

> Sometimes, in good faith, one may think that this does not represent an actual threat to the experiment.

Coping with Researcher Bias. The interviewees suggested seven strategies to cope with RB. First, the use of *pre-registration* in SE conferences/journals (see Section II-B). This should prevent negative-results papers from being rejected. Also, pre-registration increases both credibility of study results and study replicability [15]. Therefore, researchers should be less prone to bias their results. On this point, R5 said:

> Personally, I have an idea. It doesn't relate to the experimental design, rather to a discipline. It consists of having dedicated tracks of a conference or sections of a journal where authors don't submit the results of an experiment,but the experiment they plan to carry out.

Second, fostering *open data policies* in SE conferences/journals. This means not only making the gathered data publicly available, but also the analysis scripts of the study. Such open data policies should allow the reviewers (and any other researcher) to repeat the data analysis of that study so attributing credibility to study outcomes and increasing the replicability of the study. Therefore, researchers should be discouraged from biasing their studies. R1's thought follows:

> Another thing could be publishing all the analyses together with the data. But then that implies during the review process that, as a reviewer, I have to go and take a look at the analysis as well.

Third, *duplicate data analysis*. That is, two researchers analyze the same data with their own scripts without interacting with one another. Then they exchange the scripts and data to cross-check them. Finally, the results of the data analysis are com-

pared. R5 mentioned this kind of data analysis, (she/he was using at the time of the interview), which should mitigate the unconscious bias of researchers involved in the data analysis.

> The only thing I do, from about three years, is that data is always analyzed independently by two researchers. Next, they exchange the scripts and cross-check them. They exchange the data and cross-check them as well. Finally, they compare their conclusions.

Fourth, *means for increasing the awareness* of RB in SE. For example, panels on RB in SE, an ethical code for SE warning researchers against this kind of bias, or papers on RB in SE. Therefore, by increasing the awareness of RB, researchers should be warned against this kind of bias. R6 told:

> Fostering panels and discussions on this [researcher bias], conducting surveys and studies, like the one you are conducting, to understand thestatus of the community.

Fifth, *guidelines for reviewers* in SE conferences/journals. These guidelines should instruct the reviewers not to judge papers on the basis of the study results (*i.e.,* positive/negative results). As a consequence, researchers would bias the study results less because having a paper reporting positive/negative results would be equally valid. On this matter, R4 said:

> Perhaps review guidelines may also help, in the sense that you instructthe reviewers, specifically not to bias their reviews only if the results are favorable to the hypothesis of the researchers.

Sixth, *ad-hoc research tracks* in SE conferences (or ad-hoc issues in SE journals). For example, specific tracks for papers reporting negative results or specific tracks for studies having a not as strong empirical assessment. Such kind of tracksshould lead researchers not to bias their results to have more publishable results. On this point, R7 said:

> Having various publication-levels where non-rigorous studies carried out by research groups or companies can be published in prestigious journals.

Seventh, *replicated experiments* because the more the results of a study are confirmed by replications, the lower the likeli- hood of RB is. In this respect, R8 told us:

> I trust when the results are confirmed by more studies carried out by researchers that are not co-authors. I don't think only one paper is enough. I don't confide in the results of only one paper. Of course, this doesn't mean that single studies are conducted incorrectly or are error-prone, it simply impacts on generalizability.

Besides the strategies, mentioned by the interviewees, to cope with researches bias, we asked their thoughts on two further strategies, *i.e.,* blind data extraction and blind data analysis, used alone or together. In the following subsections, we report the findings concerning the sub-themes for blind data extraction, blind data analysis, and both these strategies.

1) Blind Data Extraction: Two sub-themes were defined for this theme (see Figure 1): usefulness and drawbacks of blind data extraction in SE experiments.

Usefulness of Blind Data Extraction in SE. It emerged from the interviews that blind data extraction could be a useful technique to mitigate RB because, even when extracting the metrics, a researcher could favor a given treatment based on her expectations. In other words, if the data extractor (*i.e.,*the person who is responsible of extracting the metrics from the raw dataset) is aware of research design elements (*e.g.,* treatment assignment), then the likelihood of influencing the results towards a given treatment is higher. This is why having blinded extractors would lessen the likelihood of influencing the results. In this respect, R3 said:

> Yeah, I think it [blind data extraction] sounds like a good idea. I believe that they [the researchers] may apply bad practices of statistical analysis but actually I believe more that one does it, consciously or unconsciously, while they code the data, or do it even before running the experiments because the researcher knows what treatment is and what the control is. I think that's a good idea that labels are removed and someone else transforms the data.

Drawbacks of Blind Data Extraction. As for the drawbacks of blind data extraction, the interviewees pointed out that the implementation of blind data extraction requires at least two people: an individual (*i.e.,* the study executor) responsibleof executing the experiment and another individual (*i.e.,* the data extractor) with the necessary skills to extract the metrics from the raw dataset. The latter has to be blinded to research design elements. This seems to be less feasible when both study executor and data extractor belong to the same research group—guessing or finding out about hidden information (*e.g.,* research hypotheses) would be more likely when both executor and extractor belong to the same research group. Therefore, to implement blind data extraction, it is preferable to have: *(i)* a research collaboration between two research groups where the experimenter and the extractor are not part of the same group, or *(ii)* an external expert that takes care of the metric extraction. To this respect, R8 stated:

> I think it [blind data extraction]'s complicated. In many cases it's you and your PhD student, do you really think that your student isn't aware of whodid certain things? [...] Maybe it can work in a joint experiment whereyou have a large group of people collaborating from various independent research groups. On the other hand, within the same group it is applicable in theory because you have several researchers involved, however it becomesan "open secret" as everyone is aware of what is going on. How much would it work within the same group?

Note that R5 had already used blind data extraction. She involved some experts to extract metrics from a raw dataset:

> Well now that you have mentioned it [blind data extraction], we actually have done it on two papers in the past that I had forgotten about. What we did was to gather the artifacts produced by the participants and then give all to external people who evaluated the artifacts. [...] Yes, I think this is surely useful.

2) Blind Data Analysis: Two sub-themes were defined for this theme (see Figure 1): usefulness and drawbacks of blind data analysis in SE experiments.

Usefulness of Blind Data Analysis. Blind data analysis seems a useful technique to mitigate RB. A blinded analyst (*i.e.,* an analyst unaware of research design elements) would perform the data analysis more objectively than an analyst aware of research design elements. In this respect, R7 said:

> It can be a means for a more objective analysis because it's human to be inclined to one's proposals and expectations. This can be thus an involuntary contribution, either positive or negative, that a researcher provides.

Drawbacks of Blind Data Analysis. The drawback of blind data analysis is that at least two researchers are needed—the former one conducts the study and sanitized the dataset, while the latter one performs the data analysis on the sanitized dataset. Moreover, it is preferable that the researchers do not belong to the same research groups. For example, R8 said:

This strategy also has its drawbacks. As pointed out by R5,

> It's similar to blind data extraction. That is, if you are conducting a joint experiment, you can apply blind data analysis.

3) Blind Data Extraction and Analysis: We defined three sub-themes for this theme: effectiveness of blind data analysis and extraction in coping with RB, strategies to foster the adoption of blind data analysis and extraction in SE experiments, and intention to use blind data analysis and extraction.

Effectiveness of Blind Data Extraction and Analysis. RB could arise even if blind data extraction and analysis are applied together. That is, using both blind data analysis and extraction is considered a way to mitigate RB. In fact, RB could arise not only during the metric extraction and analysis phases but also during the execution of the experiment itself. In this respect, we report R3's answer when we asked if the combination of blind data extraction and blind data analysis was enough to cope with RB:

> Most likely not. Like I said previously, the step before where you set up and where you run the experiment also introduces some [bias].

Fostering Blind Data Extraction and Analysis. The interviewees suggested a number of strategies to ease the adoption of blind data extraction and analysis in SE. The first strategy is a *policy* for conferences/journals similar to the double-blind peer-review one. That is, this policy would consist of requiring that any submitted experiment to that conference/journal had to use blind data extraction and analysis. However, this strategy is not always feasible, as the same interviewees observed, due to the following reasons: *(i)* the reviewers cannot make sure the authors of a paper have really used blind data extraction and analysis; *(ii)* researchers, who are not involved in research collaborations, would be harmed by this policy; and *(iii)* empirical evidence on the effectiveness of blind data extraction and analysis in SE is necessary to foster conferences/journals to adopt this policy. Regarding point *(ii)*, R8 said:

> In most cases, you have a [research] group that works independently... it does not involve several units, or you have a group made up of Ph.D. student and supervisor. In this case, how do you distinguish the roles and introduce any blinding in the process?

As for the last point, R4 said:

> The conference committees won't do it [that policy] without any evidence that it's gonna be effective, just because it sounds like a good idea. Then, if there is enough evidence that it's a good idea, then maybe some conferences will start using it [that policy].

The second strategy to foster the use of blind data extraction and analysis is a *third-party service provider* that takes care of metric extraction and data analysis blindly. For example, the researchers conduct the experiment and, when needed, sanitize the raw dataset (*e.g.,* it removes any label to the treatments). Then they submit the raw dataset to this service provider, which extracts the metrics and then analyzes the data. After analyzing the data, the service provider sends the results to the researches. In this respect, R5 said:

> An example could be an online service for data analysis where each participant, at the end of the [experimental] task, uploads its data on that platform and then someone else performs the data analysis. So who carries out the experiment does not interact with or manipulate the data, rather only acknowledges the results of the analysis. Clearly, this is costly and not easy to be realized.

it is not easy to realize such a system. Also, the researchers should trust the service provider as well as the people that perform blindly the data extraction and analysis. Furthermore, it would most likely introduce extra costs. The third strategy consists of a *guideline* for applying blind data extraction and analysis in SE. R6 told us:

> Someone should try to give guidelines on how to put them [blind dataextraction and analysis] in practice.

Finally, *empirical evidence* on the effectiveness of blind data extraction and analysis in SE would foster the adoption of these blind techniques. On this matter, R4 said:

> It would be nice if there could be some pilots or meta-studies that demonstrate how blind analysis and extraction change the results in either way, in favor or against the researcher's hypothesis.

Intention to Use Blind Data Extraction and Analysis. The interviewees stated they would take into account blind data extraction and analysis for their experiments. R8 stated:

> If I have to participate in a large joint experiment between several research groups, I can take this into account when assigning the roles, why not! Instead of doing everything myself.

V. Discussion and Conclusion

According to the interviewees, post-hoc hypotheses are notquestionable as long as the researchers explicitly mentiontheir use or it is possible to ground such hypotheses on priorwork. Furthermore, this practice could be used to gain newinsights into the investigated phenomenon.

Similarly, the post-hoc outlier removal practice is not always questionable. Inparticular, it is considered acceptable when the researchersprovide the results (after and before the outlier removal),justify the outlier removal, and discuss the causes behindpossible differences. This is in contrast with the guidelines forevaluating SE experiments by Kitchenham *et al.* [20] for which *"the analysis protocol needs to address how drops out werehandled"*. According to the authors, a clear outlier dropoutanalysis is particularly relevant for researchers interested inintegrating the results of similar experiments (*e.g.,* meta-analysis). In other words, from SE researchers' perspective,some QRPs are acceptable in certain cases, as recognized inprevious studies (*e.g.,* [12]). The question that arises is to whatextent QRPs relates to the presence of RB in SE experiments.Based on our findings and those by Jorgensen *et al.* [5] and Shepperd *et al.* [4], it seems that RB affects SE experiments.Thus, we need to find solutions to mitigate RB as much aspossible. Our results represent an initial exploratory step toestablish guidelines to mitigate RB in SE experiments basedon solutions for SE researches and editorial/program boards. **Solutions for Researchers**. Researchers can take into ac- count blinding techniques to extract and analyze the data oftheir experiments. The importance of applying these tech-niques is central when performing meta-analyses [20]. The researchers we interviewed were favorable to use them (or at least to take them into account) in their future experiments. Although they acknowledged the usefulness of blind data extraction and analysis, such techniques alone do not solve the problem of RB but they are means to mitigate it. Our findings suggested that blind data extraction and analysis are considered more effective in concealing information when the key roles (*e.g.,* study executor and data extractor) are covered by people that do not belong to the same research group. Therefore, we encourage researchers to *(i)* involve external experts for blind data extraction and analysis or *(ii)* collaborate with other research groups to have external researchers taking care of data extraction and analysis. However, it has emerged that involving external experts or collaborating with other research groups is not always possible. Nevertheless, it is still possible to apply blind data extraction and analysis within the same research group. For example, a simple form of blind data analysis can be achieved by relabelling the treatment groups in the dataset with non-identifying terms to hide the actual treatments from the data analyst. We recognize that in this case the analyst could guess the hidden information, but such a solution is surely better than having no blinding at all. To mitigate RB, the researcher could also consider using duplicate data analysis—*i.e.,* asking two or more people to analyze the data independently. This technique could be easily extended to data extraction. Duplicate data extraction and analysis could be applied as alternatives or in conjunction to blind data extraction and analysis. Other two solutions to counteract RB, on the researcher side, are: *(i)* replicated experiments and *(ii)* means for improving the awareness of RB. Regarding the former, the underlying assumption of the interviewees is: the more the experimental results are confirmed, the lower the likelihood of RB is. As for the latter solution, it is importantto share knowledge on RB as well as strategies to deal with it—this paper represents a first step towards this direction.

Solutions for Editorial/Program Boards. The interviewees suggested fostering open data policies to mitigate RB since themore the studies are reproducible (*e.g.,* because datasets and analysis scripts are publicly available), the less the likelihood of biasing the results is. In this respect, some conferences,such as ESEM 2018, have explicitly promoted open data policies. According to the interviewees, RB could be due tothe behavior of some reviewers, namely *(i)* their tendency to reject negative-results papers (*i.e.,* publication bias) or *(ii)* their tendency to focus too much on empirical assessment at the expense of contributions to the body of knowledge. Accordingly, reviewers' behavior can lead researchers to bias their results to make their papers more publishable. Therefore,acting on reviewers' behavior would possibly mitigate RB. The interviewees suggested ad-hoc tracks/issues for papersreporting negative results and studies having a weak empiricalassessment but with a significant contribution to the bodyof knowledge. For example, SANER 2018 has had a track where authors could submit negative-results papers, while the short paper track of EASE 2020 is fostering the submissionsof research where a weak design could invalidate interesting findings. Tracks/issues for pre-registration papers is another solution to counteract the tendency to

reject negative-results papers—*e.g.,* MSR 2020 is going to accept submissions of pre-registration papers. Guidelines for reviewers can help mit-igating publication bias as well. For example, these guidelines

should instruct reviewers not to judge papers based on study results (*e.g.,* positive/negative results). Accordingly, editorial board should enforce reviews to comply with the guidelines. Finally, conferences should consider having panels on RB in their programs to increase awareness of SE researchers about this problem and share solutions on how to limit it.

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Read introduction and answer the following questions:

1. What is researcher bias (RB), and how can it influence empirical studies?
2. What are Questionable Research Practices (QRPs), and how do they relate to RB?
3. How does publication bias impact the dissemination of research findings?
4. What are the two techniques mentioned in the paper to counteract RB, and how do they work?
5. Why are blind data extraction and blind data analysis less commonly used in Software Engineering (SE) research compared to other fields?

Read methodology section and answer the following:

6. What is the main goal of this study?
7. Why did the researchers choose interviews over questionnaires for data collection?
8. How many experts participated in the interviews?
9. What was the purpose of the "Warm Up" section in the interviews?
10. What types of biases related to RB were investigated in this study?
11. Which two techniques were the focus of the study to counteract RB?
12. How was the anonymity of the participants ensured?
13. What is template analysis, and why was it used in this study?
14. What tool was used to assist in thematic analysis?
15. How did the researchers modify their initial analysis template during the study?

Read Discussion and conclusion section, and answer the following questions:

16. What is the main objective of the discussion and conclusion section in a research paper?
17. How does this paper propose mitigating researcher bias (RB) in software engineering experiments?
18. What are the two main categories of solutions suggested in the paper to address RB, and how do they differ?
19. Why does the paper emphasize the importance of blind data extraction and analysis, and what are some challenges associated with implementing these techniques?
20. What role do editorial and program boards play in reducing research bias, according to the interviewees?

RESEARCH

Research can be defined as ***the search for knowledge***, or as any ***systematic investigation***, with ***an open mind***, to establish novel facts, solve new or existing problems, prove new ideas, or develop new theories.

Think of the word "research" as an acronym.

R
E
S
E
A
R
C
H

The word re-search is a noun composed of **two syllables: re** is a prefix, meaning again, anew or over again **search** is verb, meaning to examine closely and carefully, to test and try, or to probe Together they form a noun describing a careful, systematic, patient study and investigation in some field of knowledge, undertaken to establish facts or principles.

Diligent and systematic inquiry or investigation in an area, with the objective of discovering or revising facts, theories, applications. The goal is to discover and disseminate new knowledge. [Merriam-Webster]

Systematic investigative process employed to increase or revise current knowledge by discovering new facts. It is divided into two general categories: (1) Basic research is inquiry aimed at increasing scientific knowledge, and (2) Applied research is effort aimed at using basic research for solving problems or developing new processes, products, or techniques. [Business Dictionary]

Careful study of a given subject, field, or problem, undertaken to discover facts or principles. [The Free Dictionary]

The Traditional Scientific Method:

The traditional scientific method consists of the following activities:

Observation of selected parts of nature -----Explanation and Critical analysis of the findings------Formulation of hypothesis-------Verification through experimentation------- Fact/Knowledge generated.

Purposes of a Research:

Generally, the aim of a research is the advancement of human knowledge. Specifically, a formal research may have one of the following purposes:

1. To discover facts and ideas not previously known
2. To test existing theories and explanations for various issues and subjects
3. To achieve better and more complete understanding of something
4. To find causes behind specific occurrences and events
5. To discover new techniques and methods for things if existing one are not yieldingsatisfactory results
6. To improvise and redefine a product to increase sales, and many more……

Qualities of a Good Researcher:

- Curious
- Probing and a detective
- Critical thinker
- Creative
- Disciplined
- Hardworking and courageous
- Determined and strong-willed
- Intuitive
- Sincere and honest
- Risk taking

Importance of Research:

Research is the pumping heart of any field of study and inquiry. Crafts, techniques, technology, methodologies: everything becomes outdated and obsolete with the passage of time. The reason for this is man's thirst and passion for more information, knowledge, and improvisation; in short, research. Continuous researches and experimentations are realized as innovation and modernization. Therefore, societies lacking research culture simply fail to keep up with the pace of developed countries in the race of advancement and progress.

Research Methods:

1. **Exploratory** research structures and identifies new problems. This is done to improve the basic knowledge on the concept and walk in to the unknown realms of the subject. It is a type of research conducted for a problem that has not been clearly defined. It should draw definitive conclusions only with extreme caution. Given its fundamental nature, exploratory research often concludes that a perceived problem does not actually exist.

Here are some examples of exploratory research in computer science:

1. **Quantum Computing Algorithms**: Researchers explore new quantum algorithms to determine their potential advantages over classical algorithms. Since quantum computing is still in its early stages, the study focuses on understanding its fundamental principles rather than drawing definitive conclusions.
2. **Bias and Fairness in AI Models**: An investigation into how biases emerge in AI decision-making models and what factors contribute to them. Since the field of AI ethics is still evolving, this research helps to define the problem and suggest preliminary methods for bias mitigation.
3. **Neuromorphic Computing**: Exploring how biologically inspired computing architectures (like brain-like chips) could enhance machine learning efficiency. This research helps define potential advantages and challenges but does not necessarily offer a final solution.

4. **Security Vulnerabilities in Emerging Technologies**: Investigating how blockchain or IoT devices might introduce unforeseen security risks. The research identifies potential issues without necessarily proposing specific solutions.
5. **Explainability in Deep Learning Models**: Researchers study how and why deep neural networks make certain decisions, aiming to improve transparency and interpretability. The research is exploratory because it seeks to uncover unknown aspects rather than provide conclusive results.

2. **Constructive** research develops solutions to a specific persisting problem. This is done by technical professionals to find a new solution to a specific persisting problem. It is very commonly used in computer science research. The term "construct" is often used in this context to refer to the new contribution being developed, such as a new theory, algorithm, model, software, or a framework. This approach demands a form of validation that doesn't need to be quite as empirically based as in other types of research. Nevertheless the conclusions have to be objectively argued and defined. This may involve evaluating the "construct" analytically against some predefined criteria or performing some benchmark tests with the prototype.

Here are some examples of constructive research in computer science:

1. **Development of a New AI Optimization Algorithm**: Researchers develop a novel **genetic algorithm** to optimize neural network training for faster convergence and improved accuracy. The algorithm is tested against standard benchmarks to validate its performance.
2. **Design of a Secure Blockchain-Based Voting System**: A new **blockchain-based voting system** is proposed to enhance transparency and prevent fraud in digital elections. The security of the system is analyzed using cryptographic proof techniques.
3. **Creation of an Energy-Efficient Routing Protocol for IoT Devices**: A new **routing protocol** is designed to reduce power consumption in wireless sensor networks. The protocol is validated through simulations and compared with existing protocols in terms of efficiency.
4. **Development of an Explainable AI Model for Healthcare Diagnosis**: A novel **explainable deep learning framework** is designed to make AI-based medical diagnoses more interpretable for doctors. The framework is evaluated using real-world medical datasets.
5. **Building a Quantum Cryptography Protocol for Secure Communication**: A new **quantum cryptographic protocol** is proposed to enhance data security against quantum attacks. The protocol is mathematically analyzed and tested through simulations.
6. **A Framework for Detecting Fake News Using NLP and Machine Learning**: A new **framework combining NLP techniques and ML algorithms** is developed to detect misinformation. The model is tested on datasets of real and fake news articles to measure accuracy.

3. **Empirical** research tests the feasibility of a solution using empirical evidence. "Empirical" comes from the Greek word for experience: ἐμπειρία (empeir´ια) Empirical research is a way of gaining knowledge by means of direct and indirect observation or experience. Empirical evidence/observations can be analyzed quantitatively or qualitatively. Through quantifying the evidence or making sense of it in qualitative form, a researcher can answer empirical questions, which should be clearly defined and answerable with the evidence collected (usually called data). Research design varies by field and by the question being investigated. A combination of qualitative and quantitative analysis is often used to better answer questions.
   Here are some examples of **empirical research** in **computer science**:

1. **Performance Evaluation of Machine Learning Models for Fraud Detection**: Researchers collect **real-world financial transaction data**, apply different **machine learning algorithms**, and analyze their accuracy, precision, and recall to determine which model performs best for fraud detection.
2. **Usability Study of a New User Interface for Mobile Apps**: A study where **participants interact with a newly designed UI**, and their **click rates, task completion times, and user feedback** are recorded and analyzed to determine usability improvements.
3. **Effectiveness of Cybersecurity Measures Against Phishing Attacks**: Researchers conduct **penetration testing and surveys** on employees to measure their ability to detect phishing emails. The collected data is then analyzed to assess the effectiveness of training programs and security policies.
4. **Benchmarking Cloud Computing Performance Across Different Providers**: Researchers deploy the same workloads on **AWS, Google Cloud, and Microsoft Azure**, then **measure execution time, cost, and resource utilization** to compare their performance.
5. **Analysis of Software Development Productivity Based on Agile Methodologies**: Data is collected from multiple software development teams using Agile methodologies (e.g., **Scrum, Kanban**), and their **velocity, defect rates, and project completion times** are analyzed to determine which approach improves productivity.
6. **Measuring the Impact of AI-Powered Code Assistants on Developer Productivity**: A study where **developers use AI-based code assistants like GitHub Copilot**, and their **coding speed, error rates, and efficiency** are analyzed compared to traditional development methods.
7. **Empirical Study on the Energy Consumption of Different Sorting Algorithms**: Researchers measure the **CPU and memory usage** of various sorting algorithms (e.g., **Quicksort, Merge Sort, Heap Sort**) on different hardware configurations to evaluate their energy efficiency.

Research Techniques

Interpretivist or qualitative research techniques: Descriptive/interpretive research: empirical observation is subjected to limited formal rigor. Controls over the researcher's intuition include self-examination of the researcher's own pre-suppositions and biases, cycles of additional data collection and analysis, and peer review; Focus group research: gathering of a group of people, commonly members of the public affected by a technology or application, to discuss a topic. Its purpose is to surface aspects, impacts and implications that are of concern. Action research: the researcher plays an active role in the object of study, e.g. by acting as a change-agent in relation to the process being researched. Ethnographic research: applies insights from social and cultural anthropology to the direct observation of behaviour.

Research techniques at the scientific/ interpretivist boundary: Forecasting: involves the application of regression and time-series techniques, in order to extrapolate trends from past data. Field experimentation and quasi- experimental designs: opportunities are sought in the real-world which enable many factors, which would otherwise confound the results, to be isolated, or controlled for. Laboratory experimentation: this involves the creation of an artificial environment, in order to isolate and control for potentially confounding variables.

Quantitative and scientific research techniques: Origin in the natural sciences – scientific method Systematic empirical investigation of quantitative properties and phenomena and their relationships. The goal is develop models, theories, and hypotheses pertaining to natural phenomena (how it works) The research is generally driven by hypotheses, which are formulated and tested rigorously. Measurement is fundamental since it gives the connection between observation and the formalization of the model, theory and hypothesis Repeatability of the experiments and testing of hypotheses are vital to the reliability of the

results, since they offer multiple opportunities for scrutinising the finding.

Non-empirical techniques: Conceptual research: opinion and speculation, comprising philosophical or 'armchair' analysis and argumentative/dialectic analysis. Theorem proof: applies formal methods to mathematical abstractions in order to demonstrate that, within a tightly defined model, a specific relationship exists among elements of that model. Futures research, scenario-building, and game- or role-playing: individuals interact in order to generate new ideas, gather new insights into relationships among variables, and postulating possible, probable, and preferable futures. Review of existing literature, or 'meta-analysis': the opinions and speculations of theorists, the research methods adopted by empirical researchers, the reports of the outcomes of empirical research, and materials prepared for purposes other than research.

Engineering research techniques: Construction: involves the conception, design and creation (or 'prototyping') of an artifact and/or technique. The new technology is designed to intervene in some setting, or to enable some function to be performed. The design is usually based upon a body of theory Artifact/technology is usually subjected to some form of testing, in order to establish the extent to which it achieves its aims. Destruction: new information is generated concerning the characteristics of an existing class of technologies. Typically achieved through testing the technology, or applying it in new ways. The design is usually based upon a body of theory.

Research Paradigms in CS

Empirical: Computer science is concerned with the study of a class of phenomena. **Example:** The Structure of the Web Web does not have an engineered architecture: hundreds of billions of pages created by billions of users. Web contains a large strongly connected core (each page can reach every other). The shortest path from one page in the core to another involves 16–20 links (a small world). Analysis of web structure led to better search engines (e.g., Google PageRank method) or content filtering tools.

Mathematical: Computer Science is concerned with the study of algorithms and properties of information structures (abstraction from real objects). **Example:** Study of algorithms (Knuth) Design and analysis of (optimal) algorithms for particular problems Computational complexity Study of representation, transformation and interpretation of information structures Models for characterizing general-purpose tools Mechanisms and notations for computing all computable functions.
Engineering: managing the cost-effective design and construction of complex software-hardware systems (commercially and socially valuable) **Example:** Building a robot for the new mission to Mars And showing that it works (better than the previous model)

STEPS IN CONDUCTING A SYSTEMATIC SCIENTIFIC RESEARCH

STEP 1: SEARCHING FOR A PROBLEM TO INVESTIGATE:

In this context, the word 'Problem' means an issue, a question, a problem, a technique, a difficulty,etc which a researcher aims to study and investigate thoroughly.

Our earliest ancestors were intrigued by questions such as,

- What are the lights in the sky?
- What makes the grass grow?
- How deep is the sea?
- How far is the moon?

The basics of 'new' science are found in primitive people's ability to identify and articulate the unknowns of life and in their crude attempts to find solutions to such questions. Civilization grew as humans observed the phenomena of their world, recognized and isolated their problems, investigated them, and arrived at answers.

Therefore, a systematic scientific research begins with the recognition and selection of a problemto investigate. After selection, the problem needs to be stated in the form of a clear, guiding, and controlling statement or question. Dewey, a famous research expert, maintains that, "A problem well put is half solved".

The problem helps researchers decide what direction they must take, what is relevant and what isnot, what methodology will work, etc. In this way, the problem guides them in reaching the correctanswer to the question.
Here are some guidelines to help you select and define your problem:

1. Is the problem important or significant? Is the question worthwhile for the expenditure oftime, energy, and funds involved? What is its societal relevance?
2. Can the problem be stated in question form? Finding the answer then becomes the objectiveof the study.
3. Can the problem be delimited and narrowed? Can boundaries be defined?
4. Are resources of information available and the state of the art practical?
5. Does it interest you?
6. Do you have the required background to undertake the inquiry under question?

There are three types of problems:

1. **Problems of Fact**: these problems seek answers to what the facts are. For example, 'Is the earth flat or round?'
2. **Problems of Value**: Problems of value are involved in setting up standards or criteria. Standards of safety, health efficiency, tolerance, economy, etc within a particular situationcan be some examples.
3. **Problems of Technique**: Such problems concern the methods for accomplishing a desired result, like, 'How can a space station be launched in the mesosphere?'

Sometimes researches select a problem themselves, where as, mostly, they are hired or assigned the problem to investigate by some authority.

Hypothesis: A hypothesis is a tentative solution or answer to the problem. It is just a working guess. Scientists test their hypothesis by experiments. As a result of experiments, a hypothesis maybe proved, disproved, revised, etc.

STEP 2: DETERMINING THE PURPOSE OF THE RESEARCH:

The problem can be stated as the issue which you want to investigate. On the other hand, the purpose is defined as the reason why the investigation is undertaken. It seeks to answer the following questions?

1. What is the purpose of this research?
2. Who needs the answers?
3. Who will use the answers?
4. What is the benefit of this research?
5. What is the limitation and scope of the research?

Write a specific purpose statement to avoid deviation from the research objective.Study the example below:
Problem: What are the dietary needs of the elderly in nursing homes?
Purpose: To determine the dietary needs of the elderly in nursing homes so as to better satisfy andtreat them.
Exercise: For the following research problems, try to generate specific purpose statements.

1. How can heat loss be prevented in a two-storey brick house?
2. How are writing skills taught in the secondary section of government schools?
3. Why did the coliseum roof collapse?
4. How can the overeating of non-vegetarian food affect a person's spirituality?
5. What are the factors behind the increasing criminal activities in Karachi?

STEP 3: GATHERING BACKROUND INFORMATION ON YOUR TOPIC:

With purpose and research problem narrowed and finalized, the next most important step is to gather all information that already exists on the subject under study. For this, refer only to established material. You will report this information in the chapter titled "Review of Literature"or "Background" in your report.

This step is very important because a researcher can not make a worthwhile contribution if s/he iscompletely ignorant of what is already known about the problem. A good literature study can sometimes even yield answers to a research question. Literature review is done to achieve two goals: to find already available information about the problem, and to acquire a good backgroundin the area of research.

Literature on any subject can be found in libraries, digital libraries, automated data and informationsystems. Browse through the catalog of holdings in a library. Type of texts you will read would be

1. Standard Encyclopedia
2. Handbooks (Compact reference manuals containing the state-of-the-art data and information on particular subjects)
3. Standard Subject Textbooks
4. Periodicals (Read through the abstracts of researches)
5. Research reports, surveys, case studies, research papers, conference presentations
6. Established websites

7. Articles written by reliable writers in magazines, etc
8. Interviews of experts
9. While conducting literature search, take notes to record findings (material, facts, opinions, historical background, etc). Also, write the names of the books, journals, etc from where you are collecting information. Remember to write the title, author, page number, year of publication, and publication house. This is important to avoid plagiarism.

STEP 4: PREPARING A RESEARCH DESIGN:

Research design is the military plan of attack you make to conduct your research. It includes the following:

1. Information and data collection methodology/tools
2. A plan or procedure with tasks sequentially and logically arranged

A schedule of tasks with deadline Your research purpose, problem under study, resources, time allotted, and the report's word limit are the factors which will help you prepare a suitable research design. Let's study different data and information collection tools in detail.

a. OBSERVATION:

A common and traditional means for conducting research is through direct observation-a cornerstone of the scientific method. Observation is a serious and careful examination of object orissue under study. Follow the instructions below for a good observation to get the required data:

1. Select venue, and obtain permission from the owner.
2. Arrange access for an appropriate length of time.
3. Carry a notebook, camera, tape recorder with you.
4. Have a clear conception of the phenomena to be observed.
5. Try to ignore any preconceived notions or opinions you have in order to be objective andhonest.
6. Record what you perceive. Also, describe in detail what you find.

b. EVALUATION/CRITICAL ANALYSIS:

Evaluation is systematic determination of merit, worth, and significance of something or someoneusing authentic criteria (a set of standards).

c. EXPERIMENTING:

In experiment, events are made to occur under known conditions. Follow the instructions belowfor conducting experiments if required in your research:

1. Work out the procedure of the experiment.
2. Make sure you are aware of any precautionary measures that are to be taken.
3. Get the apparatus, equipment, devices, and other things that you will require.
4. Conduct the experiment.
5. Jot down the findings

d. INTERVIEWS:

There are two types of research interviews:

1. *Structured Interview:* A structured interview (also known as a standardized interview or a researcher-administered survey) is a quantitative research method commonly employed insurvey research. The aim of this approach is to ensure that each interview is presented withexactly the same questions in the same order. Structured interviews are a means of collecting data for a statistical survey. A structured interview also standardizes the order in

which questions are asked of survey respondents, so the questions are always answered within the same context. It generates objective numerical data.

2. *Semi-Structured Interview:* A semi-structured interview is flexible, allowing new questionsto be brought up during the interview as a result of what the interviewee says. The interviewer in a semi-structured interview generally has a framework of themes to be explored.

Here are some points to consider for a successful interview:

1. Select the right person(s). Find out his/her position/designation.
2. Make a definite appointment by telephone, letter, or email.
3. Now, prepare for the interview by formulating the right questions so that you get the desiredinformation.
4. Tape the interview. If not possible, then take notes.
5. Arrive on time.
6. Do only as much talking as is necessary.
7. If the interviewee gets off the subject, be ready with a question that can get him/her backon track.

e. QUESTIONNAIRES:

Questionnaires require judicious handling. It should be designed to require as little time aspossible for completion. Some guidelines for designing suitable questionnaires are:

1. First jot the questions you require.
2. Then phrase questions as short sentences. Avoid very lengthy questions.
3. Choose the write words to be clear and precise.
4. The first few questions should be easy so that you can secure cooperation of theparticipants.
5. One question should stimulate interest in the other.
6. Questions should be designed in such a way that a yes, no, one-word answer, ticking anoption, etc would be enough to answer it.

For questions, whether used in an interview or a questionnaire, the following is important tounderstand. There are different type of questions:

1. ***Closed Questions:*** A closed question usually receives a single word or very short, factualanswer. For example, "Are you thirsty?" The answer is "Yes" or "No"; "Where do you live?" The answer is generally the name of your town or your address. Closed questions are good for:
2. ***Open Questions:*** Open questions elicit longer answers. They usually begin with what,why, how. An open question asks the respondent for his or her knowledge, opinion or feelings. "Tell me" and "describe" can also be used in the same way as open questions.Here are some examples:

- What happened at the meeting?
- Why did he react that way?
- How was the party?
- Tell me what happened next.
- Describe the circumstances in more detail.

3. ***Funnel Questions:*** This technique involves starting with general questions, and then homing in on a point in each answer, and asking more and more detail at each level. It'soften used by detectives taking a statement from a witness:

"How many people were involved in the fight?""*About ten*."
"Were they kids or adults?""*Mostly kids.*"
"What sort of ages were they?""*About fourteen or fifteen.*"
"Were any of them wearing anything distinctive?""*Yes, several of them had red baseball caps on*."

4. ***Leading Questions:*** Leading questions try to lead the respondent to your way of thinking.They can do this in several ways:

- With an assumption: "How late do you think that the project will deliver?". This assumesthat the project will certainly not be completed on time.

Leading questions are good for:

- Getting the answer you want but leaving the other person feeling that they have had achoice.
- Closing a sale: "If that answers all of your questions, shall we agree a price?"

STEP 5: SYSTEMATIZING, ANALYSING, AND INTERPRETING THEDATA:

The data you have obtained from your investigation now needs to be systematized, organized, andinterpreted to find answers to your research problem.

How to Systematize Your Data: Arrange and organize your data by grouping it into different parts.Each part will bear information to a particular element or aspect of your problem. Give headings to each part of the data, and sub-headings if required.

How to Analyze and Interpret Your Data: Analysis and Interpretation will require you to use arange of critical thinking skills like:

1. Examining
2. Analyzing
3. Conceptualizing
4. Defining
5. Inferring
6. Understanding facts, opinions, assumptions, etc
7. Evaluating

Study each part of the data stepwise and record your findings. By this time you will have found the answer to your research problem. Now it's time to report your findings to the world. So we move to the next step which is as challenging as the research itself.

STEP 6: WRITE A REPORT TO SHARE THE RESULTS WITH THE WORLD:

The next chapter discusses report writing in great detail.

WRITING TECHNICAL REPORTS
A Reconstruction of an Investigation

A technical report is a document that describes the process, progress, or results of technical or scientific research or the state of a technical or scientific research problem.

A report is written to communicate factual and objective information obtained after a careful scientific research or scrutiny. It is usually written for a well-defined audience. The audiences require this document for an important well-defined purpose.

Importance of a Report

- Reports are written for specific individuals because these individuals need the information for future actions.
- Reports add to the body of knowledge by conveying important opinions and findings which then serve as a basis for further progress and advancement.

Types of Reports

1. **Research Report:** It communicates the results and the procedure of scientific investigation.
2. **Project Report:** It provides details about a project undertaken to solve a problem by investing a software or hardware.
3. **Feasibility Report:** It tells whether a project is feasible-that is, whether it is practical or financially and technologically possible.
4. **Recommendation Report:** It studies a problem or opportunity and then makes a recommendation. It may compare two or more alternatives and recommend one.
5. **Laboratory Report:** It documents laboratory procedures and their results.
6. **Periodic, Progress, and Trip Report:** It communicates progress on projects and events to those concerned with their completion and success. Readers review, and may revise, scheduled courses of action as a result of information included in these reports.
7. **Personnel Evaluation Report:** It reviews the performance of employees.

Elements/Components of a Report with a Standard Organizational Pattern:

COVER MATERIALS:

- Cover and title page
- Cover letter or memo (optional)
- Invoice (optional)
- Letter of Transmittal
- Abstract
- Acknowledgements
- Executive Summary
- Table of contents
- List of figures or illustrations
- List of tables

BEGINNING OF THE REPORT:

- Introduction:
  - The purpose of the investigation
  - The research problem (its nature)
  - Significance of investigating the problem

- Scope or limitation of the research
- A list of personnel engaged in the research with a brief sketch of their background and duties (optional)
- Organization of the report
- Historical Background (Review of Literature):
- Background information on the problem
- Known facts, opinions are discussed
- Nomenclature, definitions are given for new or unusual terms or those having a specialized meaning
- A list of symbols, acronyms, and abbreviations

In small scale research reports, all the items above are written in a single chapter with clear defining headings. However, in large scale research reports, 'Introduction' and 'Historical Background' are written as two separate chapters as there would be a lot of information and details to convey.

BODY OF THE REPORT:

- Discussion of research methodology (for a small scale research, this would be very brief)
- Analysis of data, results, findings, Discussion of results

ENDING OF THE REPORT:

- Conclusions
- Recommendations and alternatives (optional)
- References
- Bibliography
- Appendix
- Raw data (Questionnaires, interview questions, checklists)
- Notes taken during observations
- Rough results of experiments
- Tables or charts, photographs, etc

Writing Effective Project/Research Reports

What is a Project Report?

A project report is a detailed explanation of the project you've completed (generally in a science, engineering, or business-related course.). The project report is the formal record of the entire process you undertook to complete your Personal Project.

What is the Purpose of the Project Report?

The purpose of the project is, in the context of the degree you are studying, to integrate various aspects of the taught material and to demonstrate your (academic) research skills and your (professional) analysis, design and implementation skills. It gives you the opportunity to conduct in-depth work on a substantial problem to show individual creativity and originality, to apply where appropriate knowledge, skills and techniques taught throughout the degree programme to further oral and written communication skills, and to practise investigative, problem-solving, management and other transferable skills. The management and execution of the project is your responsibility, but you should seek and take advantage of advice from your supervisor.

When you choose a project, you should do so carefully, to reflect the focus of the degree programme you are enrolled in, your personal interests (the project needs to keep you interested for the whole the academic year) and the ability of the academic staff to support you throughout your project. Projects vary widely in the problem they address and the products they deliver at the end. While the main product of some projects is a piece of software or hardware, other projects produce a systems model or design, and yet others may address some research hypothesis using a theoretical or experimental approach. This means not every project produces a piece of software. In brief, the better defined the problem that your project addresses, the further through the systems lifecycle you should expect to progress in the course of your project. If instead you are addressing a research hypothesis, your main product may be the evaluation of some experiments or a theoretical result.

So, for example, a project that seeks to develop a logistics planning system for a small business or voluntary organisation would be expected to provide a fully operational, fully tested program that meets all the identified needs of the client. However, a project that aims to validate a government policy in a particular area might only achieve the development of a model to confidently simulate the main factors influencing that policy, and identify the research agenda in terms of specifying precisely the data requirements to allow a full investigation of the relevant factors. A scientifically oriented project may focus on the practical or theoretical evaluation of a new rendering approach and compare it with existing approaches, which may involve some implementation, but does not require fully functional software.

In short, the project report should contain information on:

- Your specific topic and goal
- Your inquiry question and the product or outcome you created
- Your Global Context
- Your plans and the process you implemented
- The resources you used
- The techniques you applied to complete the project and product
- The challenges and issues you faces and perhaps solved
- What you learned about your topic, global context
- What you learned about yourself as a learner from the Personal Project experience

How can I write and Structure a Project/Research Report?

Approaches to Writing

- Top-Down Approach
- Evolutionary Delivery

Note: You can adopt one of them, and you can use both!

Top-Down Approach

- Use Chapter Breakdown Structure to identify the structure of the report
    - Identify all chapter names, sections and sub-sections
- Identifying the contents of each chapter, makes writing much easier
- You concentrate on a certain target in each chapter, and you don't misdirect to another target
- Helps in time management

Evolutionary Delivery

You write separate parts of your reports as the thoughts come by.

- You can re-write these parts as your project proceeds, and your information increase.
- So, each part evolves and matures over a period of time as new ideas emerge.

These two approaches (Top-Down and Evolutionary Delivery) can be combined by:

- Specifying the chapters, sections and sub-sections heading and contents.
- By, taking the previous point as your road-map, you can start writing these parts, taking into consideration any probability of re-writing that might be needed while the project in progress.

How do I get the Content for my Project Report?

Most of the necessary material will consist of your own ideas and experiences gained while carrying out the project, and your approach to solving the problem you have decided to address. For the background study or literature review you will also need references to various resources such as key books and papers, policy documents, Internet resources, related software, etc.
While working on the project you may find it helpful to keep a notebook handy and record all relevant information. Typically, such information will include:

- references such as papers, books, websites with full bibliography details;
- lessons learned, for inclusion in the "reflective" part of your report;
- notes from meetings or interviews with
  - ✓ your supervisor;
  - ✓ potential end-users and other stakeholders;
  - ✓ technical experts;

Also, we recommend that you keep a diary of all your project-related activities. This will show the progress made during the life of the project and will provide a record of how you spent your time. In particular, when you are validating, testing and debugging your work, keep a running log of your activities and their outcomes. You will then have a record of the unforeseen difficulties you met and, hopefully, how you resolved them. Summaries of these may well be worth including in the project report.

In general, you should supplement the material you generate yourself with relevant material from other sources. A good project report will show that you are aware of relevant work that other people have done (see Section 3.2). You should include relevant references to such work in your project report. References to work in periodicals, i.e. magazines and journals, and conference proceedings *may* be more useful than references to textbooks, as periodicals and conferences are usually more specialised and up to date. References to technical manuals and national and international standards should also be included, where appropriate. You may also cite web sites as sources, if suitable. However, keep in mind that web sites may often contain incomplete or wrong information and in general textbooks or papers are a better reference and show that you have done a more extensive literature review than just searching for some keywords on the Internet.

What will be the Format of the Project Report?

Suggested report structure for a project which implements a piece of software	Suggested report structure for a project addressing a "softer" problem
Title Page Abstract Acknowledgements Table of Contents Table of Figures 1. Introduction 2. Background 3. Specification 4. Design 5. Implementation 6. Results and Evaluation 7. Future Work 8. Conclusions 9. Reflection on Learning References Glossary Table of Abbreviations Appendices	Title Page Abstract Acknowledgements Table of Contents Table of Figures 1. Introduction 2. Background 3. Selection of Approach 4. Application of Selected Approach 5. “Deliverables” from Selected Approach 6. Results and Evaluation 7. Future Work 8. Conclusions 9. Reflection on Learning References Glossary Table of Abbreviations Appendices

Suggested report structure for comparing algorithms	Suggested report structure for the design and analysis of an algorithm
Title Page Abstract Acknowledgements Table of Contents Table of Figures 1. Introduction 2. Background 3. Description of Algorithms 4. Implementation 5. Experiment Design 6. Algorithm Comparison Results 7. Future Work 8. Conclusions 9. Reflection on Learning References Glossary Table of Abbreviations Appendices	Title Page Abstract Acknowledgements Table of Contents Table of Figures 1. Introduction 2. Background 3. Problem Statement 4. Alternative Designs and Final Algorithm 5. Implementation 6. Experimental and Theoretical Results 7. Future Work 8. Conclusions 9. Reflection on Learning References Glossary Table of Abbreviations Appendices

How can I write the Different Sections of the Project Report?

I. Writing the Introduction

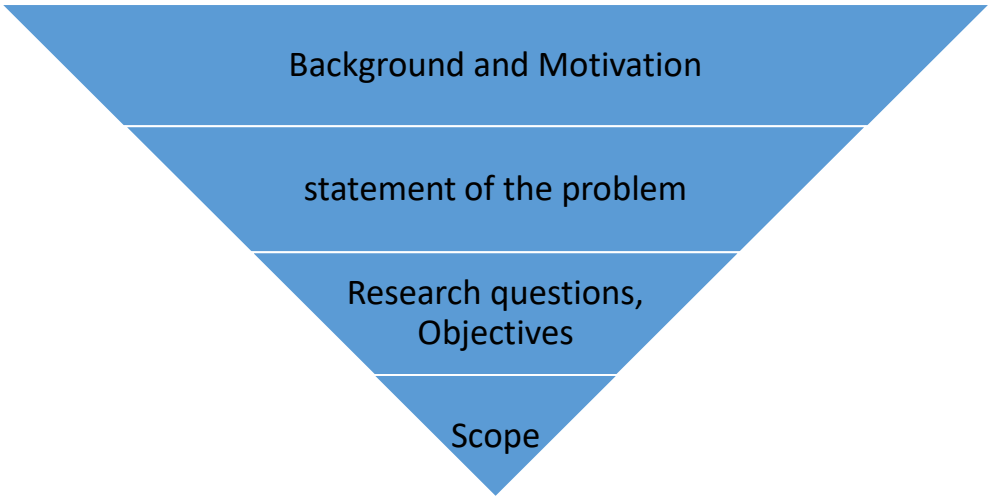
A good introduction should tell the reader what the project is about without assuming special knowledge and without introducing any specific material that might obscure the overview. It should anticipate and combine main points described in more detail in the rest of the project report. Also, importantly, it should enthuse the reader about the project, to encourage them to read the whole report. Normally it should include such things as:

1. **Why the area is important**
2. **Giving background information**
3. **Reviewing previous research**
4. **Identifying a problem or a need**
5. **Establishing a specific problem or need**
6. **End with statement of the problem**

- **The Project (You will have to make headings of the following items)**

    1. **Project description**
    2. **Purposes, aims or objectives**
    3. **Justification or importance of the project**
    4. **Project category**
    5. **Scope**
    6. **Planning & Project Management**
    7. **Outline of the structure of the report**

The flow of writing for the introduction section for an undergraduate research paper/project report:



Background and Motivation

The thoughts and actions of a researcher must be driven by some source of motivation. The motivating factors to take up a research problem can be the drive to (but not limited to);

Factors of Motivation	
- improve an existing work/project/ solution,	- build something new
- provide insights and solutions to true world problem	- deep interest in the field
- contribute to the societal, scientific, and industrial advancements	- gain new knowledge

A motivation statement is a part of the introduction section in a research paper/ project report, and it is usually provided in the lines following the background of the research. If required by the institutional template/publisher, "Motivation" may sometimes require a separate heading, under the main heading of Introduction section.

It is a broad statement that focuses on setting a stage before writing the specific/ precise problem that the research aims to solve. It may include brief discussion on the following points:

✓ **Broad context of the problem**

Start with a brief overview of the broader context your research falls into. This sets the stage for understanding the narrowed, specific problem you are addressing.

✓ **Significance of the Problem for the researcher/industry/stakeholders**

Start with a brief overview of the broader context your research falls into. This sets the stage for understanding the narrowed, specific problem you are addressing.

✓ **Current Challenges and Gaps**

Briefly analyze the effects of the problem on industry, society, academia or any other stakeholders. Discuss why it is important to solve the problem.

✓ **Specific Relevance to Your Research**

Narrow down to the specific problem your research addresses and explain why it is important to solve it.

✓ **Potential Impact of solving the problem**

Discuss the potential benefits and impact of your research. This can be in terms of technological advancements, practical applications, or contributions to knowledge.

Note: This part may also include a discussion on potential threats if the problem goes unsolved.

Sample #1

The motivation for this study is driven by the widening gap between traditional Computer Science (CS) pedagogy and the engagement requirements of modern undergraduate students. While the global software industry demands graduates with high-level persistence and deep technical mastery, traditional instructional methods often fail to sustain interest in abstract domains like computational theory and data structures. **This leads to "cognitive passivity," where students may fulfill academic requirements but lack the intrinsic drive to master complex systems.**

Addressing this engagement crisis is vital for academic and industrial stakeholders. Current challenges, such as high attrition rates in STEM and a widening technical "skills gap," suggest that failing to modernize learning frameworks could diminish the quality of the future engineering workforce. This research seeks to provide insights into how gamified mechanics—specifically real-time feedback and achievement hierarchies—can be integrated into CS curricula. By building a bridge between Human-Computer Interaction (HCI) and pedagogy, this study aims to contribute to scientific advancement and offer a practical solution for fostering industrially competent software engineers.

Commented [ni1]: Broader context: This sets the stage by identifying the general field (CS Pedagogy) and the high-level conflict (Traditional methods vs. Modern student needs).

Commented [ni2]: Significance of the problem: This explains why the problem matters to stakeholders (the industry) and identifies the specific academic area at risk (abstract domains).

Commented [ni3]: Current challenges and gaps: This analyzes the negative effects (attrition, skills gap) and the threat to the future workforce if the problem remains unsolved.

Commented [ni4]: Specific relevance: The text narrows down from the general problem of "engagement" to the specific solution being tested: gamified mechanics like hierarchies and feedback.

Commented [ni5]: Potential impact: This outlines the dual benefits: a scientific contribution to the HCI field and a practical benefit for the software industry.

Data-based technologies have lately become a common factor across industries. Efficient data retrieval is important for timely and accurate information access in fields such as healthcare, finance, and e-commerce. Despite significant advancements in SQL query optimization techniques, existing methods often struggle with execution time and resource utilization, particularly in large-scale distributed database systems. Improving SQL query execution times in distributed databases can lead to more efficient data management and better performance, which is critical for organizations handling large volumes of data. By developing a novel optimization technique for SQL queries, this research aims to enhance system performance, leading to faster data retrieval and more efficient resource use. This can significantly benefit industries relying on large-scale data processing

Commented [ni6]: Broader context

Commented [ni7]: Current challenges and gaps

Commented [ni8]: Significance of the problem

Commented [ni9]: Specific relevance to your research

Commented [ni10]: Potential impact

Activity: Now identify all parts of the motivation:

In most universities, student attendance identification is now done manually using paper, pencil, and other antiquated tools where students write or sign their names next to them. This technique is time-consuming and unproductive since it is sometimes difficult to update student records, store and maintain such information, and calculate attendance percentages for test reasons. The mechanism used to identify student attendance does not track individual attendance records over time [14]. Furthermore, since the essential data for these records are collected so that the motivation for retaining student attendance records is undermined, data from these records are often fraudulent, inaccurate, and not legitimate. False attendance and impersonation are made extremely simple due to the present procedures utilized for obtaining and preserving data on student attendance. Students find it very easy to put their friends' identities on the attendance sheet without the teacher or any other official in charge detecting it.

STATEMENT OF THE PROBLEM

To excel in engineering and computer science, you must master the art of the Problem Statement. This is not merely a complaint; it is a high-precision communication tool that justifies the existence of your project.

A problem statement is a vital communication tool used to define technical challenges and guide project development. It provides a comprehensive view of an issue by identifying its impact, location, and urgency. Beyond defining the "gap," it clarifies expected outcomes and aligns stakeholders on project goals, serving as an essential roadmap for informed decision-making and effective solution design.

Crafting the Problem Statement

A Problem Statement is a concise summary of an issue requiring timely action. In technical disciplines, it serves as the "bridge" between a functional ideal and the current problematic reality. Its primary purpose is to contextualize the issue, justify the research, and set clear objectives.

Structure: The "Three-Step" Logical Flow

To write a professional statement, follow this objective sequence:

Step 1: Contextualize the problem

Practical research problems

For practical research, focus on the concrete details of the situation:

- Where and when does the problem arise?
- Who does the problem affect?
- What attempts have been made to solve the problem?

Example: Voter turnout in the Southeast has been decreasing steadily over the past ten years, in contrast to other areas of the country. According to surveys conducted by local nonprofits, turnout is lowest among those under 25 years of age. There have been some effective attempts at engaging these groups in other regions, and in the last two elections, major parties increased their campaigning efforts. However, these interventions have yet to have any significant effect on turnout.

Theoretical research problems

For theoretical research, think about the scientific, social, geographical and/or historical background:

- What is already known about the problem?
- Is the problem limited to a certain time period or geographical area?
- How has the problem been defined and debated in the scholarly literature?

Example: In the past ten years, the "gig economy" has become an increasingly important segment of the labor market. People under 30 are now more likely to engage in freelance arrangements (rather than full-time jobs) than in the past. Research on the reasons for and consequences of this shift has focused on objective measures of income, working hours, and employment conditions. However, there has been little work exploring young people's subjective experiences of the gig economy.

Step 3: Show why it matters

The problem statement should also address the relevance of the research. Why is it important that the problem is addressed? Don't worry, this doesn't mean you have to do something groundbreaking or world-changing. It's more important that the problem is researchable, feasible, and clearly addresses a relevant issue in your field.

Practical research problems

Practical research is directly relevant to a specific problem that affects an organization, institution, social group, or society more broadly. To make it clear why your research problem matters, you can ask yourself:

- What will happen if the problem is not solved?
- Who will feel the consequences?
- Does the problem have wider relevance? Are similar issues found in other contexts?

Example: Low voter turnout has been shown to have negative associations with overall civic engagement. It is becoming an area of increasing concern in many European democracies. When specific groups of citizens lack political representation, they are likely to become more excluded over time, leading to an erosion of trust in democratic institutions. Addressing this problem will have practical benefits for the Southeastern US as well, and will contribute to understanding of this widespread phenomenon.

Theoretical research problems

Sometimes theoretical issues have clear practical consequences, but sometimes their relevance is less immediately obvious. To identify why the problem matters, ask:

- How will resolving the problem advance understanding of the topic?
- What benefits will it have for future research?
- Does the problem have direct or indirect consequences for society?

Example: Literature on the gig economy characterizes these new forms of employment sometimes as a flexible, active choice and sometimes as an exploitative last resort. To gain a fuller understanding

of why young people engage in the gig economy, in-depth qualitative research is required. Focusing on workers' experiences can help develop more robust theories of flexibility in contemporary employment, as well as potentially informing future policy objectives.

Contextualize (The Ideal): Describe how the process or system *should* work.

Example: "Ideally, cloud services should provide secure data integrity and 99.9% availability for SMEs."

Identify the Gap (The Reality): Explain the exact barrier preventing that ideal state. Focus on facts, not opinions.

Example: "However, 43% of cyberattacks now target SMEs, and current default security settings fail to mitigate ransomware threats."

Show Relevance (The Consequence): Explain the impact of leaving the problem unsolved, often in terms of financial cost, safety, or efficiency.

Example: "Unresolved vulnerabilities cost SMEs an average of $200,000 per incident, often leading to total business failure."

Essential Characteristics (The "VOII" Framework)

For your problem statement to be accepted by journals or stakeholders, it must be:

Valid: Logically sound and technically grounded.
Original: Addressing a new parameter, scenario, or objective.
Impactful: Offering a broad benefit to the field or industry.
Identified (Precise): Sufficient in depth while remaining strictly within your project's scope.

The "CS/Engineering" Checklist

When drafting your statement for a manual or report, ensure you have addressed these "5 W's":

Who does the problem affect? (e.g., Software testers, end-users)
What is the specific technical failure? (e.g., Latency, resource leakage)
When/Where does it occur? (e.g., During peak traffic, in distributed nodes)
Why must it be fixed now? (e.g., Financial loss, security breach)

From Problem to Aim

Your problem statement should lead directly to your Aim (the overall purpose) and Objectives (the concrete steps).

Aim: To investigate effective engagement strategies in CS pedagogy.
Objective: Using statistical analysis, the research will measure the impact of leaderboards on student uptime.

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Example of Problem Statement: Enhancing Cyber security in Cloud Computing for Small and Medium Enterprises (SMEs)

1. How Things Should Work

Cloud computing enables businesses to store, manage, and process data efficiently while reducing IT infrastructure costs. Ideally, companies should be able to rely on cloud services without the risk of cyberattacks, unauthorized access, or data breaches. Secure authentication, encryption, and compliance with industry standards should ensure data integrity, availability, and confidentiality.

2. **The Problem and Why It Matters**

Despite its advantages, cloud computing faces critical cybersecurity vulnerabilities, particularly for Small and Medium Enterprises (SMEs) that lack advanced security infrastructure. Cyber threats such as ransomware attacks, data breaches, and insider threats have increased, leading to financial losses, reputational damage, and compliance violations. Many SMEs either lack the expertise to implement strong security measures or rely on default cloud security settings, which may be inadequate. Addressing these security gaps is essential to protect sensitive business and customer data.

3. **Financial Costs of the Problem**

According to industry reports, cyberattacks cost SMEs an average of $200,000 per incident, with 60% of affected businesses closing within six months due to financial losses. Additionally, data breaches may lead to regulatory fines, loss of customer trust, and potential lawsuits. Without adequate security, SMEs are at high risk of operational disruptions and financial instability.

4. **Supporting Evidence**

- IBM's Cost of a Data Breach Report (2023) states that cloud-related breaches cost businesses an average of $4.35 million per incident.
- A survey by Verizon (2023) found that 43% of cyberattacks target SMEs, as they are often less protected than large enterprises.
- According to Gartner, by 2025, 99% of cloud security failures will be due to human errors and misconfigurations.

5. **Proposed Solution**

To mitigate cybersecurity risks, SMEs need an AI-driven Cloud Security Framework (AI-CSF) that includes:

1. Automated Threat Detection – AI-powered tools that monitor and detect anomalous activities in real-time.
2. Zero Trust Security Model – Implementing strict authentication and authorization controls.
3. End-to-End Encryption – Encrypting data both in transit and at rest.
4. Security Awareness Training – Educating employees on phishing attacks, password management, and social engineering threats.

6. **Benefits of the Solution**

Implementing the AI-CSF will:

- Reduce cybersecurity incidents by up to 80%, lowering financial risks.
- Increase regulatory compliance, ensuring SMEs meet security standards like ISO 27001 and GDPR.
- Enhance customer trust by providing secure data handling and privacy protection.
- Improve operational efficiency through automated security monitoring.

7. **Conclusion**

Cyber security vulnerabilities in cloud computing pose a severe threat to SMEs, leading to financial losses and reputational risks. By adopting AI-driven threat detection, Zero Trust security, and strong encryption, businesses can enhance data protection and resilience against cyber threats. Investing in cloud security is not just a precaution but a necessity for business continuity and growth.

How to write a Problem Statement in Computer Science:

To clearly present a problem in the computer science research (or any domain of scientific research), the problem statement must include the following three main aspects in the statement for the audience:

I. Context in which the problem is important,
II. the precise problem
III. the solution

Sample

Despite the critical importance of student engagement in undergraduate Computer Science education, traditional lecture-based methods often result in low motivation and high attrition when tackling abstract technical concepts. Currently, there is a lack of empirical evidence regarding which specific gamified elements—such as leaderboards, digital badges, or experience points—most effectively improve long-term engagement for CS students. Many existing learning applications are "badly designed," relying on superficial rewards that fail to address the specific cognitive load of programming. Without a systematic investigation into the impact of these applications, educators continue to rely on inefficiently structured pedagogical tools, resulting in suboptimal learning outcomes and a workforce unprepared for the rigorous demands of the software industry.

Look at the following sample statements of problem.

Sample # 1

The public sentiments towards various topics has become an important factor with the growing influence of social media, products, or events. Sentiment analysis algorithms aim to automatically classify the polarity of social media posts (positive, negative, or neutral), but they often struggle to accurately capture the nuances of language and context. ***This limitation compromises the reliability and effectiveness of sentiment analysis, hindering its practical applications in areas such as market research, brand management, and public opinion monitoring.*** *To address this challenge, our project proposes the development of advanced sentiment analysis algorithms that incorporate deep learning techniques and contextual understanding. By using neural networks and natural language processing models trained on large datasets, we aim to improve the accuracy and robustness of sentiment classification.*

Sample #2
There is a growing demand for automated image analysis in Healthcare industry for medical imaging analysis, such as detecting abnormalities in X-rays, MRIs, and CT scans. To meet this demand, enhancing object detection capabilities is crucial. *This research explores how a neural network can enhance object detection in images. By using annotated images for training, the study aims to improve the accuracy and efficiency of object recognition tasks."*
Sample #3 *[READ IN CONTINUATION OF SAMPLE #1 OF MOTIVATION STATEMENT]*
Ride booking services are becoming increasingly common for daily commute in urban areas due to a variety of factors. This increases the importance for the service providers to accurately predict the ride demand for efficient operations. Current ride booking systems struggle to predict ride demand accurately, leading to inefficiencies in service allocation and resource management. *This research addresses the challenge by employing a machine learning algorithm to analyze historical booking data for improved demand forecasting. By using historical booking data as a training dataset, the study aims to develop a machine learning model that classifies and predicts ride demand patterns. This solution intends to enhance operational efficiency and customer satisfaction through better demand management.*
Sample # 4
sorting algorithm is one that puts elements of a list in a certain order. It makes easy searching and locating the information. The most-used orders are numerical order and lexicographical order. An efficient sorting algorithm is that takes less time and space complexity. In this paper, a contrastive analysis of bubble sort and merge sort is carried out, and the requirement for a new approach to get best sorting results has been argued. *In this regard we proposed a hybrid approach that will take minimum number of comparisons, with less time and space complexity to sort. Divide and conquer (merge sort) with bi-directional bubble sort approach is used and an example data is sorted with such hybrid approach.*

Exercise:

REMEMBER: Your focus in this course is on the language part, and on learning to write the research problem statement using correct flow and format. If you feel stuck on a scenario given in the exercises, invent details using your imagination without worrying about the technical correctness. *However*, when you write the actual research papers, you will need to be technically correct in your statements.

Write problem statements using the following information. Use verbs such as design/propose/develop/offers.

1. Biometric system+ biometric device+ identify students' finger prints+ uses data of registered students
2. Intelligent crime combating system+ track illegal acts+ find prepetrators
3. Computerized restaurant menu system+ automate order placement+ track order
4. Computerized census system+ automate recording, retrieval of data+ enable direct access to required data

RESEARCH OBJECTIVES

The Research Objectives should outline the specific goals that the research intends to achieve. These objectives should be clear, concise, and directly related to the problem statement. This section often includes several bullet points or numbered items, each describing a specific objective.

Conciseness:

- **Be Brief and to the Point:** State each objective in one or two sentences without unnecessary details.
- **Example:** "To develop a machine learning model that predicts ride demand based on historical booking data."

Action-Oriented Language:
Use **Active Verbs:** Start each objective with an action verb to clearly convey the intended activity.
Example **Verbs:** Develop, evaluate, design, analyze, implement, test, compare, optimize.
Relevance:
Align with the Research Problem: Each objective should directly relate to addressing the research problem.
Example: "To identify key factors affecting SQL query execution time in distributed databases."
Logical Structure:
Order Objectives Sequentially: Arrange objectives in a logical sequence, often from most to least important or in the order they will be addressed.
Example: First develop the model, then implement it, and finally test its performance.
Consistency:
Maintain Uniform Language: Use a consistent tone and style throughout the objectives.
Example: If using technical terms, apply them uniformly.

Samples

- To **develop** a novel optimization technique for SQL queries in distributed databases.
- To **evaluate** the performance of the proposed technique through benchmarking on large-scale datasets.
- To **compare** the proposed optimization technique with existing methods in terms of execution time and resource utilization.

Exercise: Given the following brief description of a research problem, write three specific research objectives.

a) **Research Problem:** Cloud computing has become a cornerstone of modern IT infrastructure, providing scalable and flexible resources. However, cybersecurity remains a significant concern, as cloud environments are vulnerable to various attacks such as data breaches, unauthorized access, and Denial of Service (DoS) attacks. Current security measures often fail to provide robust protection due to their inability to adapt to new and evolving threats. This research seeks to enhance cybersecurity in cloud computing by developing adaptive security protocols that can dynamically respond to threats.

b) **Research Problem:** Autonomous vehicles (AVs) have the potential to revolutionize transportation by reducing human error and improving traffic efficiency. However, one of the key challenges is ensuring the safety of AVs in complex and unpredictable environments. Current algorithms sometimes struggle with accurately detecting and responding to sudden changes, such as pedestrians suddenly crossing the road or unexpected obstacles. This research aims to develop advanced machine learning models to enhance the real-time decision-making capabilities of AVs.

Writing Research Scope

In research writing, **scope** defines the precise boundaries within which your study operates. It identifies the "territory" your research covers, including the specific population, timeframe, geographic location, and technical variables under investigation.

Why it matters: A well-defined scope prevents "scope creep," ensuring the project remains feasible and focused. It signals to reviewers that you understand the limitations of your resources and have intentionally narrowed your focus to produce statistically significant and valid results.

How to write it: To define your scope, explicitly state what is included and, crucially, what is excluded (delimitations). Use specific parameters:

- **Population:** Who are the subjects?
- **Duration:** What is the timeline?
- **Variables:** Which specific technical factors are being measured?

By clearly stating these boundaries, you protect the integrity of your conclusions and provide a clear roadmap for future researchers to expand upon the areas you chose to omit.

Examples of Scope Statements

1. The scope of this research is limited to undergraduate Computer Science students currently enrolled in core programming modules. The study specifically evaluates the impact of three gamified elements: progress tracking, competitive leaderboards, and achievement badges. While the study explores motivation and engagement, it will not assess the impact of gamification on final examination grades or long-term career success. Evaluation will be conducted through a mobile-based gamified learning application over a single academic semester.

2. This project will consist of creating a marketable game based upon the *Bouncy Bunny Counts Money* comics on our website. The project will be completed by December, 2010. Modules of the game will include a simple shopping game, a way for Bouncy Bunny to earn money, and a way to motivate players to continue to play.

3. This project involves building a fence between the house at 10 ABC Boulevard and 12 ABC Boulevard. The fence will consist of steel posts within concrete-filled holes. The fence will be built out of cedar and it will be 8 feet tall. This is anticipated to keep the dog at 10 ABC Boulevard contained within the yard at a reasonable cost. The fence will be located as close to the property line as possible and reach from the garage on the west side to the house on the north side.

4. This project is for the creation of a construction safety app for cell phones. There will be an app for iPhones and Android based systems. The user interface will be designed as part of the project but will contain, as a minimum, the ability to create and edit tailgate meetings, field level hazard assessments, safety inspections, and audits. Each of these will have a built-in checklist for typical projects in typical industries. There will be a corresponding web application whereby anyone using the app can log in to view and print the reports. The app must include a tutorial to make it easy to get started.

Organization

Chapter 1 gives an overview of lifelong learning and discusses the aim of this project. Chapter 2 talks about the related work done in lifelong learning and online learning. Chapter 3 explains Decision Theoretic Online learning (DTOL) and the adaptive algorithms used in online learning setting, AdaHedge and Follow the Leader with dropout perturbations (FTL-DP). Chapter 4 introduces the lifelong learning problem and explains EWA-LL and FTL-DP in lifelong learning setting. Chapter 5 explains the dataset and experimental setup. Chapter 6 gives a conclusion for the project and discusses the future work which can be carried forward from this project.

Exercise: Critically read the following "Introduction" and insert appropriate headings.

Introduction: Automatic Marking of Exam Papers Using Semantic Parsing

'Automated Essay Scoring' has been **a large area of research since the 1960s**. In such a process, a variety of 'features' are extracted from essays, such as word and sentence length and the structure of sentences, before the data is collaborated to provide a final classification **[SHERMIS 03].**

'Automated Exam Scoring' is a more objective mode of classification, in which answers are analysed for the presence of concrete facts or statements instead of using any continuous measure. **This has been the subject of much research** in the Computational Linguistics department at Oxford University, using an online study as the source of data, in which students completed a GCSE Biology paper **[PULMAN 05].** The techniques employed are varied, but fall into two main categories. One simulating a human style marker defines the marking scheme via patterns inputted by an administrator, allowing for as many variants of an answer as possible. **The clear disadvantage of this method is the hours of work required to painstakingly write these patterns**, but this method yields high accuracy. Average accuracy in excess of 95% was obtained.

The latter method adopts a machine learning approach using a set of pre-marked answers for the training process. **[PULMAN 06]** experimented with a system in which the answers are treated as a 'bag of words' with no semantic structure incorporated. A technique known as 'k nearest neighbour (KNN)' was used…

This naive method is subject to a number of problems, as highlighted by Professor Stephen Pulman…

"You can't just look for keywords, because the student might have the right keywords in the wrong configuration, or they might use keywords equivalents".

Thus if the answer requirement is a statement such as 'the cat chased the mouse', then an answer of 'the mouse chased the cat' would be accepted despite the clear semantic inequality, due to the identical set of words.

This project aims to extend this method by incorporating the semantic structure of sentences, **so that** for the above example 'the mouse chased the cat' would be marked as incorrect, whereas 'the mouse was chased by the cat' would be marked as correct.
The CAndC **(Clark and Curran)** parser uses statistical methods and 'supertagging' to convert English sentences into a tree representing their structure, **as detailed in [CLARK 07]…**The output of the parser is a CCG (Combinatory Categorial Grammar) file, representing this sentence structure.
Alone, this representation is insufficient for machine learning use, given that the semantic interpretation of the sentences is our concern. **We therefore use** a tool called Boxer, which uses Prolog to convert the CCG into a form called DRS (Discourse Representation Structure). **This is compatible with first-order logic, and thus can be used to make reasoned logical deductions** (with its application extending to other systems such as Question Answering).

Exercise: Read the following Introduction and insert headings given below:

- Assignment definition
- Claims for the project
- The purpose
- The scope and limitations of the project
- The field of study
- Existing work related to the project

Component 1	The principle of a "robot" is an ancient one. The word robot in Czech means labourer or worker. Commercial robots do not have the 'intelligence' to think independently. This project aims to address this inadequacy by improving existing robotic technology.
Component 2	Thomas Ross built the first robot mouse in the 1930s, but subsequently robotics developed at a very slow pace until the late 1970s, when useful industrial robots became a practical proposition.
Component 3	The project team had to build a computer controlled maid robot that could successfully serve drinks and food as ordered.
Component 4	The mini-robotic toy described here, which we have built and called RoboMaid, is an important step in this direction in that it can serve cold drinks and food only. The use of computer control not only greatly increases the scope and capabilities of the robot, but the machine itself takes programming out the two dimensional world of the visual display unit into the three dimensional world.
Component 5	Our objective was to produce a mini-robot which consisted of three units: the computing and processing unit, the transceiver unit and the driving unit.
Component 6	A range of programming was developed so that the robot can sense its environment via the sensing device coupled to its "shell". Although the robot is able to serve cold drinks, it works most effectively if only one type of drink is ordered.
Component 7 Outline of the process/procedures	We designed it to be controlled by 4 data lines derived from the Motorola G4 AltiVec computer processor. Binary bits are used to control the motor drive circuits. By turning on, or off, different combinations of the bits 0-4, quite a variety of movements can be executed. The action of the control will be described in the software part. We used a transceiver unit which transmitted or received infra-red and the signal was manipulated by the computing and processing unit.

Exercise: Write the Introduction section for a report using the information below:

The group developed an Educational Game called "Numbers is order" for primary school children. The game requires the user to click the bubbles as fast as possible in a specific order. Each time the order will change between all, odd, or even and increasing or decreasing.

Instructions for the Game are as follows:

Click "Tap Now" to being the game.

A number of balls will drop from the top of the screen. Each ball has a number in it.

At the bottom of the screen it will say ODD or EVEN or BLANK at the bottom left and INCREASE or DECREASE at the bottom right. You must click the balls in this order. The first ball to click is blinking.

If the bottom left is BLANK (says nothing) you click all the balls in order.

Example:

If it says ODD and INCREASE at the bottom and the first number is 1.

Then you click the bubbles 1, 3, 5, 7, 9, etc. until the level is complete.

Tip: Check the order at the bottom before you worry about the numbers in the balls.

Tip: There is a limited amount of time to finish. The timer is at the bottom center of the screen.

This game should work on all platforms including safari and mobile (we hope, but make no guarantees).

Exercise: Imagine that your group was required to develop a software program as a mandatory part of a course. Your group developed an Airline Reservation System. This system is a complete application for airline reservation programmed in VB.net. Airline Reservation System can inform clients regarding all flights, their arrival, their departure, and seat availability. Through this system, the clients can book, cancel, and print tickets according to the scheduled flights.
Write the Introduction section for this Project. You are required to write the following sections only.

1. **Motivation (150 to 200 words; follow the format taught)**
2. **Project Definition (1 to 2 sentences; follow the correct format)**

NOTE: Brainstorm to get ideas and make an outline to organize your thoughts. Do not use the internet. You can invent details and use imaginary citations. [10+3]

II) Review of Related Researches/Background

The purpose of the Background section is to provide the typical reader with information that they cannot be expected to know, but which they will need to know in order to fully understand and appreciate the rest of the report (see Section 4.1 for details of who a typical reader might be). It should explain why the project is addressing the problem described in the report, indicate an awareness of other work relevant to this problem and show clearly that the problem has not been solved by anyone else. This section may describe such things as:

- the wider context of the project;
- the problem that has been identified;
- likely stakeholders within the problem area;
- any theory associated with the problem area;
- any constraints on the approach to be adopted;
- existing solutions relevant to the problem area, and why these are unsuitable or insufficient in this particular case;
- methods and tools that your solution may be based on or use to solve the problem;
- and so on.

The wider context of the project includes such things as its non-computing aspects. So, for example, if you are producing software or any other products, including business recommendations, for a specific organisation then you should describe aspects of that organisation's business that are relevant to the project.
Relevant existing products, documents or artefacts that you should mention could be ones that, for example,

- are similar to the one you are proposing;
- support your project;
- your project aims to extend or replace;
- demonstrate the "deficiencies" your project intends to address.

You need only describe things that will be unfamiliar to the potential reader, or are unique to the organisation or topic your project addresses. Your project, if it involves software development, will almost certainly use all kinds of existing software such as language compilers, subroutine libraries, etc., but you can assume that the reader will be fully acquainted with, for example, general purpose programming languages such as Java, C/C++, Fortran, Pascal, Python, PHP, etc,. Also, it may involve the better known specialised packages such as MySQL, ORACLE, OpenGL, etc. You should mention the particular variety and possibly version number, e.g. Java SE 6, but you need say nothing more than that.

If your project depends on any specialist or uncommon software such as specialised subroutine packages or a more obscure or specialised programming language, you should describe them briefly and discuss whatever features are relevant to your project. Often this can be done by comparing it to some well-established piece of software, for example

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The Descartes language is like a restricted version of Pascal but with
the following extra features:
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Your background section should end with a clear statement of the research questions problem your project is trying to answer. These will reflect the aim of your project, but will be different in that they explain the problem you are attempting to solve, e.g.,

Example 1:

Aim:

The aim of this project is to develop software for the improved planning of the routing of delivery vehicles to customer locations, that reflects the forecast availability of each customer to receive goods.

Research question(s):

In order to demonstrate the achievement of the stated aim, this project will identify route planning software currently in use and the underpinning algorithms, define appropriate performance metrics, determine how to express constraints on an alternative algorithm, develop an improved algorithm and demonstrate on what basis it is judged an improvement, and implement the improved algorithm in a usable and robust software package.

Example 2:

Aim:

The aim of this project is to develop a business strategy for organisation X that will improve the survivability of X in the face of increasing global competition.

Research question(s):

In order to develop a business strategy, it will be necessary to identify key stakeholders and determine their vision for the organisation at the end of the strategic planning timeframe, assess the likely outcome, in

terms of the organisation's survivability, of maintaining the current strategy, and develop and assess an alternative set of activities to achieve the stated vision.

General Steps for Writing a Literature Review

Here are general steps to write a literature review.

1. **Stage One: Annotated Bibliography**. As you read articles, books, etc., on your topic, write a brief critical synopsis of each. After going through your reading list, you will have an abstract or annotation of each source you read. Later annotations are likely to include more references to other works since you will have your previous readings to compare. But at this point the goal is to get accurate critical summaries of each individual work.

2. **Stage Two: Thematic Organization**. Find common themes in the works you read and organize them into categories. Usually, each work in your review can fit into one category or sub-theme of your main theme. But sometimes a work can fit in more than one. Write some brief paragraphs outlining your categories, how in general the works in each category relate to each other, and how the categories relate to each other and to your overall theme.

3. **Stage Three: More Reading**. Based on the knowledge you have gained in your reading, you should have a better understanding of the topic and of the literature. You have discovered specific researchers who are important to the field or methodologies you were not aware of. Look for more literature by those authors, on those methodologies, etc. You may be able to set aside some less relevant areas or articles which you pursued initially. Integrate the new readings into your literature review draft. Reorganize themes and read more as appropriate.

4. **Stage Four: Write Individual Sections**. For each section, use your annotations to write a section which discusses the articles relevant to that theme. Focus your writing on the theme of that section, showing how the articles relate to each other and to the theme, *rather than* focusing your writing on each individual article. Use the articles as evidence to support your critique of the theme rather than using the theme as an angle to discuss each article individually.

5. **Stage Five: Integrate Sections**. Now that you have the thematic sections, tie them together with an introduction, conclusion, and some additions/ revisions in the sections to show how they relate to each other and to your overall theme.

The following can be the format for a literature review. You can modify it as per your needs.

1. Establishes research territory
2. Establishes significance of territory
3. Establishes research niche. (Briefly reviews the major achievements in the field)
4. Reviews the chronological development of research in this area (an approach that is useful at times, but not always the best). Discusses one key paper at a time by describing its methods and key findings, but then **identifies weaknesses in the method and/or limitations in the findings**. Then discusses how the next researchers tried to address these problems.
5. Uses the found weaknesses to justify the project topic and aim
6. End with a summary and states the research problem with force and conviction

<u>Exercise: Critically read the following Literature Review and comment on its strengths and weaknesses.</u>

1.1 Related Works.

Surface reconstruction from unorganized point samples has been extensively studied in computer graphics,

computational geometry and computer vision (see [Dey 2007] and references therein). Our main goal, however, is different. We focus on acquiring and understanding large 3D indoor environments.

Scanning technology. Powered by recent developments in realtime range scanning, everyday users can easily acquire 3D data at high frame-rates. The individual frames, however, are poor in quality. Hence, researchers have proposed algorithms to accumulate multiple scans for better quality acquisition [Henry et al. 2010; Izadi et al. 2011]. Unfortunately, such methods lead to ghosting artifacts if the camera or the scene moves abruptly in course of scanning. Furthermore, the raw scans do not provide any high-level understanding of the scene. Scan processing. Rusinkiewicz et al. [2002] first demonstrated the possibility of real-time lightweight 3D scanning. In their framework, the user rotates a handheld object while the system continuously updates the model to provide real-time visual feedback guiding the user where to scan next. In other related efforts, researchers have used template models to learn the space of human bodies [Allen et al. 2003], morphed database models to fill in regions of missing parts [Pauly et al. 2005], used non-rigid alignment to better align (warped) multiple scans [Li et al. 2009], or exploited non-local repetitions to consolidate point clouds for urban facades [Zheng et al. 2010]. The methods, however, are not suitable for extracting a high-level scene understanding along with appropriate deformation models from sparse and unorganized inputs.

Shape analysis. Man-made objects populating indoor scenes typically have low-degree of freedom and are often arrangements of simple primitives. Schnabel et al. [2007] introduce an algorithm to automatically extract basic shapes (e.g., planes, cylinders, spheres, etc.) from unorganized point clouds. Subsequently, the GlobFit [Li et al. 2011] framework extracts a set of mutually consistent relations (e.g., coplanar, coaxial, equal length, etc.) and conform to the recovered relations for reverse engineering. Alternately, temporal information across multiple frames can be used to additionally track joint information in order to recover a deformation model ([Chang and Zwicker 2011] and references therein).

In the context of image understanding, Lee et al. [2010] construct a box-based reconstruction of indoor scenes using volumetric considerations, while Gupta et al. [2010] apply geometric constraints and mechanical considerations to obtain a block-based 3D scene model. In the context of 3D scans, there has been little efforts towards scene understanding of large datasets. Notable exceptions include: Triebel et al. [2010] present an unsupervised algorithm for segmentation and object detection in indoor scenes. They apply a graph-based clustering on pre-segmented input data and assign part labels using a Conditional Random Field (CRF). The method, however, does not consider object variability and cannot be applied to unorganized pointsets, as is our goal. Boyko et al. [2011] extracts high level information of road in noisy outdoor point sets. We also extract high level information in the context of indoor environments for quick and effective scene understanding.

3. BODY OF THE REPORT:

This is the longest section of the report. It contains the following:

1. The research methodology: It contains a technical description of the research design. Second, it offers a rationale or justification for selecting the respective research methods (data collection methods). It discusses in detail the procedure with steps arranged in the right chronological order. It discusses the sampling strategy used along with a rationale for it.
2. It analyzes the data. Different elements of the data are analyzed separately by referring to the appendix for a closer look at the raw data mass. It discusses the answers that the data yield. It discusses everything that the data disclose. It presents the information in a highly organized and readable form. Use charts, graphs, tables, headings and subheadings, pictures, diagrams-anything that can ease the process of reading.
3. It utilizes many visuals (graphs, charts, diagrams, etc) to ease the presentation and explanation of key points.

III) The "Specification & Design"

The purpose of the Specification and Design sections is to give the reader a clear picture of the system you plan to create, in terms of the capability required. A specification should tell the reader what the software system is *required* to do. The design then gives the top-level details of how the software system meets the requirement. It will also identify constraints on the software solution, that are important in guiding decision making throughout the development process.

Describing what a software system does (specification) and how it does so (design) effectively usually means describing it from more than one viewpoint. Each viewpoint will convey some information about the system that other viewpoints omit. (You would use the same technique when describing any complicated construction such as a building, an aircraft, a novel or a painting). Possible viewpoints might be:

- the business model the software supports;
- the user interface;
- the dynamic behaviour of the system;
- how data flows through the system;
- what data types are implemented in the system;
- what algorithms are implemented in the system;
- the static architecture of the system, i.e. how the code is partitioned into modules, etc.

A common approach is to first define the user or business requirements, then describe the static architecture, identify modules and groups of closely connected modules, and then to apply other views to each of these groups. Fine details, specifically details of code, should be left out.
We strongly recommend that you make extensive use of diagrams, such as entity-relationship diagrams, UML diagrams, state charts, or other pictorial techniques.

As well as describing the system, it is important that you *justify* its design, for example, by discussing the implications of constraints on your solution and different design choices, and then giving reasons for making the choices you did. Typically these implications will relate to the aims of the project and to aspects of it discussed in the Background section.

The design of the system will almost certainly have evolved while you were developing it. Obviously you should describe its final state but often there are good reasons for describing intermediate states, too; for example, if you want to discuss the details of the design method used or to highlight learning that you later refer to in the Reflection section. If you do this, take special care to make sure the reader does not get confused between different stages of the design.

If you are not designing a system, but testing a hypothesis for a more scientifically oriented project, specification and design sections may not be required in quite the same form. The specification instead becomes a description of the problem and what is required of a solution. The design becomes a description of your approach to solving the

problem and your suggested soltuion(s). For instance, if you are designing an algorithm to solve a particular problem you would have a problem statement section and then a section describing one or more suggested algorithms to solve the problem. Later in the Results and Evaluation section you then describe how to design experiments to test how well the algorithm(s) solve the problem and present your experimental results with an evaluation of your suggested solutions.

IV) The "Implementation"

The Implementation section is similar to the Specification and Design section in that it describes the system, but it does so at a finer level of detail, down to the code level. This section is about the realisation of the concepts and ideas developed earlier. It can also describe any problems that may have arisen during implementation and how you dealt with them.

Do *not* attempt to describe all the code in the system, and do *not* include large pieces of code in this section. Complete source code should be provided separately. Instead pick out and describe just the pieces of code which, for example:

- are especially critical to the operation of the system;
- you feel might be of particular interest to the reader for some reason;
- illustrate a non-standard or innovative way of implementing an algorithm, data structure, etc.

You should also mention any unforeseen problems you encountered when implementing the system and how and to what extent you overcame them. Common problems are:

- difficulties involving existing software, because of, e.g.,
- its complexity,
- lack of documentation;
- lack of suitable supporting software;
- over-ambitious project aims.

A seemingly disproportionate amount of project time can be taken up in dealing with such problems. The Implementation section gives you the opportunity to show where that time has gone.

V) The "Results and Evaluation"

In this section you should describe to what extent you achieved your goals. You should describe how you demonstrated that the system works as intended (or not, as the case may be). Include comprehensible summaries of the results of all critical tests that were carried out. You might not have had the time to carry out any full rigorous tests – you may not even got as far as producing a testable system. However, you should try to indicate how confident you are about whatever you have produced, and also suggest what tests would be required to gain further confidence.

This is also the place to describe the reasoning behind the tests to evaluate your results, what tests to execute, what the results show and why to execute these tests. It may also contain a discussion of how you are designing your experiments to verify the hypothesis of a more scientifically oreinted project. E.g., describe how you compare the performance of your algorithm to other algorithms to indicate better performance and why this is a sound approach. Then summarise the results of the tests or experiments.

Exercise 2: Critically read the Requirements section of a report.

3D Modelling in Java

The 3D modelling system required for this project **must be cleanly accessible** from within our Java code, allow for dynamic changes to the 3D world, and **provide a high-level intuitive interface** for doing so. **What we require is** a system which can interface cleanly with the Eclipse window, and allow user-interaction with the underlying 3D objects.

One such three-dimensional modelling language **satisfying these requirements** is Java3D. **The reason for this is that** it provides a way to create a three-dimensional scene, completely in Java, and in a high-level manner…
In designing any program, one must consider the requirements, in terms of fulfilling and achieving certain goals, whilst also adhering to the requirements in efficiency and usability enforced by an end-user. **I will now discuss what these requirements are:**

Accuracy… Efficiency… Usability… Extensibility… Integration…

This list prescribes themes which should feature throughout the design process, whilst giving an overview of what we plan on achieving. **We will now continue to describe** various aspects of the design which aims to meet these requirements.

Exercise 3: Critically read the Design and Testing section of a report.

Design
4.1 Preliminaries
This section will discuss the methods used in setting up a framework to allow for the dynamic placement of 3D visual objects.
4.1.1 Creating the Eclipse Plug-in
Creating an Eclipse plug-in is a straightforward process. Dave Springgay gives a good outline of the processes necessary [13]. However, essentially we are concerned with creating an Eclipse 'View'…
4.5.5 A Different Approach to Determining Object Size
As we have seen in the Divide and Resize algorithm, the visual objects size can play a vital role in the usability of the general layout. The halving method employed in the divide and resize algorithm **seems rather naive**, even if it works well visually. Given that the model has access to an importance score for each object, **it would seem nonsensical** for two objects to be of the same size, when one is vastly more important than the other. **Hence, I suggest a sizing algorithm based solely on the importance of the object…**
Testing
We have already seen some screen shots of the working program; however, we provide two stringent tests for our program to ensure it works as intended, along with a test rig to fully analyse the program. In both test programs, I will run through the whole series of options available to the user, and ensure its correctness.

However, I will also demonstrate its ability to visualise code, and hopefully provide valuable insights whilst debugging.

5.1 Simple Program -BFS and DFS using the Visitor Pattern

This test program begins by creating an underlying tree structure…

This kind of debugging is intuitive, and simple to do within this framework. If you have an intuitive understanding of what the underlying model in your program should look like, it is fairly straight forward to spot bugs like this in small code samples. Assuming a larger program is in use, the user must delve a little deeper into the part of the graph which they suspect the bug to exist in. This is obviously heavily aided by the JDT debugger itself. However, this test still shows the usability of the code in a small program, and shows that the code can cope with the different types of back links and cross links that can occur in a memory graph.

VI) The "Conclusions"

The Conclusions section should be a summary of the aims of project and a restatement of its main results, i.e. what has been learnt and what it has achieved. An effective set of conclusions should not introduce new material. Instead it should briefly draw out, summarise, combine and reiterate the main points that have been made in the body of the project report and present opinions based on them.

The Conclusions section marks the end of the project report proper. Be honest and objective in your conclusions. You must also critically evaluate your results in the light of these tests, describing its strengths and weaknesses. Ideas for improving it can be carried over into the Future Work section. Remember: no project is perfect, and even a project that has failed to deliver what was intended can achieve a good pass mark, if it is clear that you have learned from the mistakes and difficulties. This section also gives you an opportunity to present a critical appraisal of the project as a whole. This could include, for example, whether the methodology you have chosen and the programming language used were appropriate.

VII) The "Future Work"

It is quite likely that by the end of your project you will not have achieved all that you planned at the start; and in any case, your ideas will have grown during the course of the project beyond what you could hope to do within the available time. The Future Work section is for expressing your unrealised ideas. It is a way of recording that „I have thought about this", and it is also a way of stating what you would like to have done if only you had not run out of time. A good Future Work section should provide a starting point for someone else to continue the work which you have begun.

Exercise 4: Critically read the Conclusion section of a report.

Efficient Local Type Inference

I have successfully developed a novel local type inference algorithm from an initial specification of the problem. **I first provide** an intuitive derivation of the algorithm **and then offer** a formal proof of correctness. **I go on to offer** generalizations to the algorithm, **supporting more** language features, and **finally achieve** local type inference in Jimple.
I have carried out careful experimental evaluation to compare the performance of my algorithm to that of Gagnon et al. [2], which is the only implemented alternative for local type inference in JimpIe. **Experiments showed a typical 4-fold to 5-fold execution time improvement across a wide range of benchmarks, and a much greater increase** where very large methods exist. **Experiments were not confined to** code compiled from Java, and include code compiled from very different languages like Scala and Scheme. **My algorithm is proven to always give a tightest possible typing.** Experiments show that Gagnon's algorithm rarely but sometimes gives suboptimal typings.

The theoretical worst case complexity of my algorithm is exponential: whereas Gagnon's algorithm is polynomial. But experiments of execution time against method length show a typically linear trend whereas Gagnon's show a cubic trend. My algorithm is very much optimized for type hierarchies in which most pairs of types have a single least-common-ancestor, which probably includes most Java bytecode in existence today. **Of course if a language appeared that made much greater use of multiple inheritance then my algorithm may not be appropriate. But as a practical 'workhorse' implementation I believe this is seriously worthy of consideration. And this is supported by** the decision of the Soot framework's maintainers to replace their existing type inference algorithm with mine.

7.1 Future Work

The greatest scope for future work is extending the application of my algorithm from local type inference to global type inference. This involves inferring types for method signatures and public fields as well as local variables. **The global type inference problem is currently an area of active research,** most of which builds upon the work of Palsberg and Schwartzbach [6]. **One could begin by** treating method parameters, return values and fields in the same way as local variables, and then using my algorithm on the program as a whole.

THE CONCLUSION OF A REPORT

The concluding section of the report usually comprises three components as outlined below. Notice that the first component looks back at what has been done and the last two componentslook ahead or beyond your project using the new experience gained from your project.

1) Meeting of objectives

Here you are required to tell the reader whether you met your objectives, and if not, the extent towhich you did meet them. If there was any change in your initial objectives you must give the reasons for this change.

2) Proposing/Recommending Design improvements to your project

Having worked on your project for some time, you may have thought of other ideas that couldimprove your piece of work in terms of its cost, efficiency to design, etc. These should be indicated at this juncture.

Remember that when you are making recommendations you are saying what *should* be done toremedy a situation or to improve something. Look at the example below:

The following recommendations are made:
4.1 *(the correct no)* the feasibility of using perspex *should be* considered;
4.2 the base *should be* attached with super glue instead of screws;
(Appropriate no e.g. 4) Recommendations:
It is recommended that :
4.1 (the correct no.) the feasibility of using perspex be considered;
4.2 the base be attached with Superglue instead of screws;
The above use of be by itself is the use of the subjunctive i.e. we use just "be" instead of *should be*.
When you make suggestions you are no longer saying what should be done; you are offering ideas about what could be done.
Suggestions:
1. the base of the sharpener could be extended to give it greater stability;
2. the sharpener could be made in different colours to make it more attractive.

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Suggestions:
1. the base of the sharpener could be extended to give it greater stability;
2. the sharpener could be made in different colours to make it more attractive.

3) Suggesting New Lines of Study
Here you are required to move away from the confines of your own project and tell the reader if the ideas contained in your work could be applied elsewhere, giving rise to new assignments, projects and new areas of study.

EXERCISE: CONCLUSION OF REPORTS:
Look at the content of the two conclusions A and B below taken from a project report and answer the following questions.

1. In what way are they similar?
2. In what way are they different?
3. Study the use of passive voice.
4. What modal verbs are used to give suggestions and recommendations?

Conclusion A :

A robot called RoboMaid **was successfully built** by our team.

The robotic toy could , **however, be further improved** to do other tasks **by** using more sophisticated software and a wider range of programs.

Using the same principle, future projects should undertake the construction of other applianceswhich could perform more useful tasks than just serving drinks.

Conclusion B :
The Automatic Firewall and Routing Apple OS X server **was successfully completed** by our team. Most of our objectives were met. This project **was also nominated** one of the three outstanding awards of Temasek Polytechnic, School of Engineering Project Show 2000. We were able to build a machine that was able to access 5000 iMacs in a flexible and user-friendlymanner. The machine is fully integrated and is easy to operate.

The machine could be further improved in numerous areas. The following changes **havebeen recommended** :

1) Adding memory cards to increase limited RAM space of the controller.
2) Using a G5 microprocessor to control and stabilise the current source during start up.
3) Using discrete digital selector and display to select current level.More
work has to be done in the following areas :
1) The function of the machine could be expanded to access Linux computers.
2) Another UNIX server could be integrated for security purposes.

Read the following conclusion and recommendation and critically analyze it.

This study investigates the impact of the foreign capital inflows and economic growth on stock market capitalization in 18 Asian countries by using the panel data from the period of 2000-2010. The ARDL bound testing co integration approach confirms the valid long run relationship between considered variables. Results indicate that foreign direct investment has significant negative economic growth has significant positive relationship with the stock market capitalization; whereas, the results of workers' remittances is found insignificant in long run. The error correction model confirms the significant positive relationship of economic growth workers' remittances while, FDI has negative and significant impact on stock market capitalization in short run. Results of causality test based on Toda and Yamamoto (1995) show the bidirectional causal relationship of foreign direct investment and direct economic growth with stock market capitalization. However, no causal relationship is found in between workers' remittances and stock market capitalization. It suggested that investor should not idealize the inflow of workers' remittances to invest in Asian stock markets in long run. Simultaneously, size of the economy is a better leading indicator for Asian stock markets. On the other hand, inflows of FDI may mislead the investor to invest. Investor should keep an eye whether FDI come in the competition of domestic market or not? If this happens so investor should not invest in the stock market of host country.

For the given conclusion, write recommendations.

i. Price and quality of food

While it was generally agreed that the price was reasonable and the variety wide, morethan half of those surveyed felt that the quality of food should could be improved upon particularly in the areas of the taste of the food, size of food portions and freshness of ingredients.

ii. Competition

Many canteen users, especially staff, were attracted by a new food court at the new neighborhood shopping centre opposite the polytechnic. Canteen takings were also affected by six newly installed non-carbonated health drinks vending machines locatedin the canteen. Many staff also found sandwiches from the sandwich bar located outsidelecture theatre 4 of DP tasty and healthy. Compounding the problem could be the canteen did not seem to serve enough types of healthy food.

iii. Canteen operating hours

Respondents, majority of whom were staff, were dissatisfied with the canteen operatinghours. They wished the canteen to be opened as early as 8 am to cater to those wantedto have breakfast or avoid the lunch crowd.

iv. Ambiance of the canteen

The ambiance of the canteen was not conducive for having meals due to the lack of airconditioning and high noise levels. The music played by the newly installed juke box worsens the situation as the selection of music turns away the adults. Hence, to improvethe canteen recommendation must address the problems identified in the above areas.

How to Write the Elements/ cover material of the Report

COVER & TITLE PAGE:

The title page must have the following features:

1. Neat and organized
2. Clearly indicates the subject title in the upper half of the page
3. The title of the report should specifically identify both the report's function and subject
4. The person or organization writing the report and the person or organization receiving the report, the date of submittal appear in the lower half of the page.

Look at the example below:

FEASIBILITY STUDY OF OFFICE EXPANSION
FOR UNITED COMPUTER TECHNOLOGIES, INC.

Prepared for
Joanne R. Galloway
Senior Vice President
United Computer Technologies, Inc

By
William D. Santiago
Senior Partner
PRT Management Consultants, Inc.

March 3, 1999

EXERCISE: Design a title page using the information given below:

1. Geologic Report on The Physiographical Development of the Colorado Piedmont Area
2. By Maurice De Valliere, Geology Student
3. Submitted to Dr. Herman M. Weisman, Professor of Technical Journalism, Colorado Polytechnic University, Fort Collins, CO 80521
4. November 30, 2008

LETTER OR TRANSMITTAL:

The letter of transmittal is addressed to the official who authorized the report and is signed by the official authorized to produce the report. It is the official acknowledgement of completion of the report and includes a statement of its transmittal. It contains the following information:

1. The title and subject of the report

2. Can include a very brief summary
3. It can acknowledge those who assisted in preparing the report
4. It has a formal tone.

Study the sample letter of transmittal below and try to write one for your own project.

December 9, 1982

David McMurrey, Chairman
Coastal Real-Estate Developers
400 Baywater Blvd.
Corpus Christi, Texas

Dear Mr. McMurrey:

As agreed in our September 21 contract, we are submitting the attached report entitled *The Effects of Increased Atmospheric Carbon Dioxide*.

This report examines the problem of CO2 accumulation in the earth's atmosphere. The climatic changes caused by excessive CO2 concentrations in the atmosphere, and the implications of these changes, will be discussed. Also discussed are the mechanisms of the greenhouse effect, the sources of atmospheric carbon dioxide, and some possible remedies to the problem.

I hope you find this report satisfactory.

Sincerely yours

William R. Waters, President
Environmental Research Associates, Inc.
1212 Trace Dr., Suite 3
Austin, Texas 78741

WRW: mb
Enclosures

ABSTRACT:

An abstract tells the potential reader the contents of the report. An abstract gives the gist of essentials of the investigation or the contents of the report. It may explain the problem and how it was studied. It gives enough description that would be sufficient for the reader to determine whether the report pertains to their interest or not. It has a standard word limit from 100-250 words depending on the size of the report.

Exercise: Read the abstract below and perform the tasks that follow:

1. State the research problem of the abstract in your own words.
2. What is the motivation for this study? Explain in 1 to 2 sentences only.
3. Name the games that the study worked with?
4. What are the strengths of the study as indicated by the results?

ABSTRACT

Board games applications usually offer a great user experience when running on desktop computers. Powerful high-performance processors working without energy restrictions successfully deal with the exploration of large game trees, delivering strong play to satisfy demanding users. However, nowadays, more and more game players are running these games on smartphones and tablets, where the lower computational power and limited power budget yield a much weaker play. Recent systems-on-a-chip include programmable logic tightly coupled with general-purpose processors enabling the inclusion of custom accelerators for any application to improve both performance and energy efficiency. In this paper, we analyze the benefits of partitioning the artificial intelligence of board games into software and hardware. We have chosen as case studies three popular and complex board games, Reversi, Blokus, and Connect6. The designs analyzed include hardware accelerators for board processing, which improve performance and energy efficiency by an order of magnitude leading to much stronger and battery-aware applications. The results demonstrate that the use of hardware/software codesign to develop board games allows sustaining or even improving the user experience across platforms while keeping power and energy low.

Read the following abstracts and answer the questions that follow: [10]

Abstract 1

This report summarizes what the computing research community knows about the role of trustworthy software for safety and effectiveness of medical devices. Research shows that problems in medical device software result largely from a failure to apply well-known systems engineering techniques, especially during specification of requirements and analysis of human factors. Recommendations to increase the trustworthiness of medical device software include (1) regulatory policies that specify outcome measures rather than technology, (2) collection of statistics on the role of software in medical devices, (3) establishment of open-research platforms for innovation, (4) clearer roles and responsibility for the shared burden of software, (5) clarification of the meaning of substantial equivalence for software, and (6) an increase in FDA's access to outside experts in software. This report draws upon material from research in software engineering and trustworthy computing, public FDA data, and accident reports to provide a high-level understanding of the issues surrounding the risks and benefits of medical device software.

Abstract 2

In the past, Wall Street used to be the center for most of the Stock Exchange and Brokerage firms due to the limitations of communications technology. Today, with the advancement in ICT, investors can use revolutionary Internet Client–server technology to trade stocks nearly anywhere, anytime, independent of brokers' fee and service limitations. This Project JTrade is a stock market trading program in Java that supports portfolio management, charting, technical analysis, paper trading, and experimental methods like genetic programming. It has features such as equation storing, quotes storing, graphical analysis of stock, maintaining portfolios, paper trading, etc.The application provides public low–cost transactions and cutting–edge, real–time market information that formerly belonged only to brokers. It has opened up extraordinary new investment opportunities as well as a crucial need for state–of–the–art information.

Questions:

1. **State the research problem of Abstract 2 in your own words.**

2. **According to Abstract 1, how many sources are used to collect data regarding understanding the pros and cons of medical device software?**
3. **What is the motivation/justification offered by Abstract 2 for developing JTrade? Explain in your own words.**
4. **State any two recommendations presented by Abstract 1.**
5. **Describe the functionalities of JTrade.**

Exercise: Read the following technical report and write an ABSTRACT for it within 200 to 250 words.

INTRODUCTION

With the advent of mobile network communication system, users have been offered lots of services such as ability to send multimedia messages like SMS, Video, Data files, Images etc. This paper describes the design and implementation of a system (Result Alert System) that conveniently provides examination results to students with the use of Email and SMS technology via their Mobile phones and devices. Mobile phones and devices are necessary assets, most especially to students as it makes communication and the spread of information a lot easier.

Result Alert System takes advantage of some of the technologies that Mobile devices provide, technologies such as the Email and Short Message Service (SMS). The implemented system allows registered students to access their results, including past results, provided they are available in the system's database.

OVERVIEW OF THE EXISTING WORK

Most Institutions make use of a web based platform to provide Examination results to their students when it is available; this is a common practice by various Universities. The students provide certain information before they are able to access their results as this provides protection from unauthorized access. Web based platforms (websites) are readily available platforms and can provide various options such as saving and printing. But this platform becomes inaccessible if the user has no access to the internet. Web based Result Alert System is a good option in a country where internet is readily available, but in a country where internet services are poor, it can be inconvenient and expensive. Students would have to visit a cyber café and pay to gain access to their results.

METHODOLOGY

The result alert system with email and SMS was designed to work as an online application or software. The system was designed to have a point of Entry which is to be used by the Administrator with the login privilege and role granted.

The Administrator is responsible for the following: registration of students, result upload for registered students and generating alert reports to all concerned students when new result updates are available. These alerts would be sent to the phone number the student provided at the point of registration.

This system was designed using the following: PHP programming language, PHP designer, Javascript, Css.

SYSTEM ARCHITECTURE

The diagram in Figure 1 shows the system architecture of the proposed system. Student initiates the examination results retrieval process by sending SMS to the specific number provided by the system.

The required information to be written in request SMS are Full name, Matric number, department, semester and session. This information is set as requirements for request SMS in order to suit with

the institution's regulations. This information together with the sender's mobile phone number are then sent to the GSM modem via GSM network.

RESULTS AND DISCUSSION

Figure 2 shows the screen shot of where the user (student) can provide his or her unique identifier in order to gain access to his or her results.

CONCLUSION

Result Alert System with Email and SMS is an innovative addition to the education sector, as it makes the availability of results and grades a lot easier and efficient. Also it makes Email and SMS technology relevant in Educational sector.

EXECUTIVE SUMMARY:

Executive summaries are written for busy administrators and decision makers who will not have the time to read the entire report. Also, executive summary is written to save a person from reading a report on issues or subjects in which s/he lacks the required background. Guidelines for writing a good ES are:

1. Read the entire report to grasp its full content.
2. Your summary should be one-quarter of the original text.
3. Write persuasively to motivate readers to read the report.
4. Discuss your research problem, the purpose of the study, the methodology used, the findings and results. Also, make recommendations if required.
5. Use headings to facilitate reading.
6. Be simple and formal.

Study the example ES below:

Executive Summary	
This report provides an analysis and evaluation of the current and prospective profitability, liquidity and financial stability of Outdoor Equipment Ltd.	subject matter
Methods of analysis include trend, horizontal and vertical analyses as well as ratios such as Debt, Current and Quick ratios. Other calculations include rates of return on Shareholders Equity and Total Assets and earnings per share to name a few. All calculations can be found in the appendices.	methods of analysis
Results of data analysed show that all ratios are below industry averages. In particular, comparative performance is poor in the areas of profit margins, liquidity, credit control, and inventory management.	Findings
The report finds the prospects of the company in its current position are not positive. The major areas of weakness require further investigation and remedial action by management.	Conclusions
Recommendations discussed include: • improving the average collection period for accounts receivable· • improving/increasing inventory turnover· • reducing prepayments and perhaps increasing inventory levels	Recommendations (note that conclusions and recommendations can be bulleted)

The report also investigates the fact that the analysis conducted has limitations. Some of the limitations include: forecasting figures are not provided nature and type of company is not known nor the current economic conditions data limitations as not enough information is provided or enough detail i.e. monthly details not known results are based on past performances not present	Limitations of the report.
Executive Summary 2 This report was commissioned to examine why the sales volume of Choice Chocolate has dropped over the past two years since its peak in 1998 and to recommend ways of increasing the volume. The research draws attention to the fact that in 1998, the market share of Choice Chocolate was 37%. The shares of their key competitors such as Venus and Bradbury were 22% and 18% respectively. The size of the chocolate market then was $36 million. Over the next two years, although Choice Chocolate retained its market share the volume of sales in the whole market decreased to $29 million. Further investigations reveal that this market shrinkage coincided with an increase in health awareness amongst consumers who regard the milk and sugar ingredients in chocolate as negative;moreover, since the second half of 1999, an increasing number of rival 'health candies' had appeared on the market. These claimed to offer the consumers a healthy alternative. These factors appear to be the major causes of the decreased sales volume of Choice Chocolate. Slim Choice is the latest chocolate range put forward by the R & D Department of Choice Chocolate. The report evaluates this range and concludes that it would be an ideal candidate to meet the challenge presented by the market and could satisfy the new consumer demand since it uses significantly reduced milk and sugar ingredients and is endorsed by renowned health experts. According to 97% of the 2000 subjects tested recently, it also retains the same flavour as the original range. It is recommended: • that Choice Chocolate take immediate measures to launch and promote Slim Choice alongside its existing product range; • that Slim Choice adopt a fresh and healthy image; • that part of the launch campaign contains product endorsement statements by renowned health experts; • that Slim Choice be available in health food shops as well as in traditional chocolate retail outlets	Terms of reference Statement of problem/ topic Formal language appropriate to report writing Key findings summarised Problem solution summarised Recommendations summarized
Executive Summary 3 Every time a business or consumer purchases products or services they display forms of buyer behaviour that are	Background to problem Report's aims

influenced by many factors. The following report looks at the fast food industry and will analyse four McDonalds' key products and services.It highlights what type of consumer buying or business buying behaviours are displayed in the purchase of a product or service and explains why each behaviour may occur. This enables a conclusion to be drawn from applying theory to reality. Although a full comprehension of buying behaviour is impossible, since everyone is an individual, it is useful to reflect on common behaviours and attempt to divide behaviours in types and stages. Even McDonalds, a leader in marketing cannot always predict consumer behaviour.	Outlines what information the report deals with but FAILS to provide a summary of the results gained, conclusions drawn and recommendations made. These are the functions of an executive summary and are absent in this example. The information in this executive summary is vague rather than summarising what the report found.

TABLE OF CONTENTS/LIST OF FIGURES/TABLES:

A well-constructed table of contents helps readers determine the subject matter of the report, its organization, and the location of sections of interest. Divide the material into major headings, subheadings, and further subheadings. Number all the headings for quick reference. Two formats for numbering headings are:

1. The traditional (I.,I.A.,I.A.1, and so on)
2. The multiple decimal format (1.0,1.1.1,1.1.2,1.2,1.2.1 and so on)

Study the example below:

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